



Verified Carbon Standard

600 MW MONSOON WIND POWER PROJECT



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1 PROJECT DETAILS

1.1 Summary Description of the Project

The Monsoon Wind Power Project is located in the Sekong and Attapeu Provinces of Lao PDR. It will be the first wind power project in Lao PDR, the largest in Southeast Asia, and the first cross-border in Asia developed under a Memorandum of Understanding between the Governments of Lao PDR and Vietnam. The project will provide a substantial source of clean renewable energy supply to Viet Nam to help meet the country's growing energy demand. The project will help unlock Lao PDR's significant untapped wind resource potential and provide social and economic benefits to the country in the form of employment, improved infrastructure and increased regional connectivity.

The project will implement cutting-edge onshore wind energy technology to achieve a total generation capacity of 600 MW, with an estimated average annual output of 1,707 GWh. The electricity generated will be supplied to Vietnam's power grid through its connection to the Thanh My substation. By harnessing the wind energy potential of the region, the project will produce clean electricity, reducing reliance on fossil fuels and contributing to regional energy security.

Before the project's implementation, the area lacked large-scale renewable energy facilities, and electricity generation heavily relied on conventional, carbon-intensive sources. This reliance resulted in limited renewable energy contributions to the grid and elevated greenhouse gas (GHG) emissions.

By replacing fossil fuel-based electricity with wind power, the project is estimated to achieve an annual average reduction of approximately 1,347,847 tCO₂e through the generation of 1,707 GWh of clean electricity. Over its operational lifetime, the project will significantly contribute to global climate change mitigation efforts by displacing substantial amounts of carbon-intensive energy from the grid.

1.2 Audit History

The project is yet to be commissioned and has not undergone validation or verification. Therefore, this section is not applicable at this stage.

Audit type	Period	Program	Validation/verification body name	Number of years
<i>Validation/verification</i>	<i>(DD-Month-YYYY–DD-Month-YYYY)</i>	VCS	<i>Validation/verification body name</i>	<i>One year</i>
	...			

1.3 Sectoral Scope and Project Type

Sectoral scope¹	1 - Energy (renewable/non-renewable)
Project activity type	Renewable Energy Project

Complete the table below with information relevant for AFOLU projects:

Sectoral scope	NA
AFOLU project category²	NA
Project activity type	NA

1.4 Project Eligibility

1.4.1 General eligibility

The Project is a 600 MW grid connected electricity generation activity using wind power in Lao PDR, which is a Least developed Country (LDC)³ as designated by the United Nations. As per VCS Standard v4.7, section 2.1 -Table 1, grid connected electricity generation activities using wind power plants located in LDCs are within the VCS program scope, making this project eligible under VCS.

As per VCS Standard v4.7, section 3.8, the project start date is the date on which the project began generating GHG emission reductions and as per subsection 3.8.1, all Non-AFOLU projects shall complete validation within two years of the project start date. This project is yet to start commercial operations and the project proponents are seeking pipeline listing and prior consideration before commencing the operations.

Developed under the CDM-approved methodology ACM0002 – Grid-connected electricity generation from renewable sources – Version 22.0⁴, this project aligns with VCS GHG program and methodology standards, qualifying it for participation in the VCS program.

1.4.2 AFOLU project eligibility

Not Applicable. This project is a wind power project which falls under the non-AFOLU project category.

1.4.3 Transfer project eligibility

¹ Projects, activities, or methodologies may be developed under any of the 16 VCS sectoral scopes: <https://verra.org/programs/verified-carbon-standard/vcs-program-details/#sectoral-scopes>

² See Appendix 1 of the VCS Standard

³ https://www.un.org/development/desa/dpad/wp-content/uploads/sites/45/publication/ldc_list.pdf

⁴ <https://cdm.unfccc.int/methodologies/DB/XB1TX7TAZ6SLWM9B7BC67THHVD16JV>

Not Applicable as this project is not registered under any other GHG programs and is not seeking to transfer.

1.5 Project Design

Indicate if the project has been designed as:

- ☒ Single location or installation
- ☐ Multiple locations or project activity instances (but not a grouped project)
- ☐ Grouped project

1.5.1 Grouped project design

Not applicable as this project is not a grouped project

1.6 Project Proponent

Organization name	Monsoon Wind Power Company Limited
Contact person	Mr. Sira Khamklai
Title	Interim O&M Manager
Address	No. 004, Unit 01, Xieng Yuen Village, Chanthabouly District, Vientiane Capital, Lao PDR
Telephone	+66 2825 9211
Email	Sira@impactelectrons.com Kanokploy@impactelectrons.com

1.7 Other Entities Involved in the Project

Organization name	Kosher Climate India Private Limited
Role in the project	Project Representative
Contact person	Vamsi Krishna M
Title	Director

Address	Zee Plaza, No.1678, 27th Main Rd, Near KLM Mall, Sector 2, HSR Layout, Bengaluru, Karnataka 560102, India
Telephone	+91 99453 43475
Email	vamsi@kosherclimate.com narendra@kosherclimate.com

1.8 Ownership

The project activity is being undertaken by Monsoon Wind Power Company Limited, it owns all the legal rights and is the sole owner of the project activity. It has already obtained all the necessary permits/consents from the various concerned government departments that are required to develop and operate a wind power project in the Lao PDR. Further, Environmental Impact Assessment (EIA) license, equipment purchase contract, and the power purchase agreements are the evidences for the ownership of the plant, equipment's and power generation. This is in line with the VCS Standard, v4.7, subsection 3.7.1 requirement 3.

1.9 Project Start Date

Project start date	31-March-2025 (Scheduled commercial operation date)
Justification	Project start date is the date on which project started generating electricity and hence began generating GHG emission reductions. This project is yet to start it's commercial operation.

1.10 Project Crediting Period

Crediting period	☒ Seven years, twice renewable ☐ Ten years, fixed ☐ Other (state the selected crediting period and justify how it conforms with the VCS Program requirements)
Start and end date of first or fixed crediting period	31-March -2025 to 30-March-2032

1.11 Project Scale and Estimated GHG Emission Reductions or Removals

- ☐ < 300,000 tCO₂e/year (project)
- ☒ ≥ 300,000 tCO₂e/year (large project)

Calendar year of crediting period	Estimated GHG emission reductions or removals (tCO ₂ e)
31-March-2025 to 31-December-2025	1,019,194
01-January-2026 to 31-December-2026	1,347,847
01-January-2027 to 31-December-2027	1,347,847
01-January-2028 to 31-December-2028	1,347,847
01-January-2029 to 31-December-2029	1,347,847
01-January-2030 to 31-December-2030	1,347,847
01-January -2031 to 31-December-2031	1,347,847
01-January-2032 to 30-March-2032	332,345
Total estimated ERRs during the first or fixed crediting period	9,438,621
Total number of years	7
Average annual ERRs	1,348,374

1.12 Description of the Project Activity

The project involves the development of a 600-megawatt (MW) wind power facility in Lao PDR, designed to export and supply electricity to neighboring Vietnam. It marks a significant milestone as the first wind power project in Lao PDR, the largest in Southeast Asia, and the first cross-border wind energy project in Asia. This initiative will provide Vietnam with a substantial source of clean, renewable energy to meet its growing energy demands while unlocking Lao PDR's vast untapped wind energy potential.

The project will bring numerous social and economic benefits to Lao PDR, including job creation, improved infrastructure, enhanced regional connectivity, and increased revenue through royalties, lease payments, and taxes. By displacing approximately 1,707,000 MWh of electricity annually from thermal fossil fuel-based power plants, the project is expected to achieve an estimated annual average reduction of 1,348,374 tCO₂e in greenhouse gas (GHG) emissions.

Beyond emission reductions and diversifying power sources, the project will create significant employment opportunities, driving economic growth in local communities. The wind power sector is expanding, with wind technicians becoming one of the fastest-growing professions in Lao PDR. Local communities near the project area will be prioritized for employment opportunities,

fostering improvements in education, lifestyle, and economic well-being. This project is set to transform local livelihoods, ensuring long-term social and economic development for the region.

Project technical specifications and the location of key Project components are set out below:

Technical Specification - Wind Turbine		
1	Wind turbine model	EN-171/4.512 and EN-171/4.5
2	Number of units	Total 133 (125 WTGs 4.512 MW and 8 WTGs 4.5 MW)
3	Rated power output	4.512 MW and 4.5 MW
4	Generator	
	Generator type	Double-fed Induction
	Nominal power	4,700 kW
	Rated Voltage	950 V
	Rated Frequency	50 Hz
5	Rotor	
	Rotor Type	Gearbox wind turbine
	Rotor Diameter	170.8 m
	Swept Area	22,965 m ²
	Rated wind speed	10.5 m/s
6	Blades	
	Blade Type	3 blades and horizontal axis
	Blade Length	83.9 m
7	Tower	
	Tower Type	Steel
	Hub Height	110 m
8	Wind turbine foundation	
	Foundation Type	Anchor cage
	Depth of foundation	Approximately 4.0 m
	Total area of each hardstand (133 units in total)	2,840 m ²
	Total volume of concrete in each foundation	Approximately 670 m ³
	Total amount of steel in each foundation	Approximately 80 tonnes excluding anchor cage and embedded plate
	Crane pad area	18 m x 25 m (Crushed stone layer)

Intermediate 35kV/115kV Substations

Each intermediate substation (4 in all) consists of several Medium Voltage (“MV”) circuit breakers for receiving power from MV feeders, 35kV/115kV transformers, and high voltage transmission facilities for outgoing feeders to the on-site step-up facility.

Key components of each Intermediate 35kV / 115kV substation		
1.	35 kV Transmission lines	
	• Type	Overhead line
	• Voltage	35 kV
	• Tower type	Hot dip galvanized shaped steel
	• Total height	26.7 - 32.7 m
	• Foundation	Extended foundation using soil, steel re-enforced and concrete
2.	115 kV Transmission lines	
	• Type	Overhead line
	• Voltage	115 kV
	• Tower type	Hot dip galvanized shaped steel
	• Total height	27.25 – 33.25 meters for single trip 33.4 – 39.4 meters for double trip
	• Foundation	Long pile, short pile and dug foundation
Key components of each Intermediate 35kV / 115kV substation		
3.	35kV / 115kV Transformer	
	• Type and configuration	3-phase, copper winding and Oil Natural Air Natural (ONAN) cooling system
	• Rated power	165 MVA
4.	MV circuit breaker	
	• Breaker type	Sulfur Hexafluoride (SF6) and Vacuum type
	• Rated maximum voltage	35 kV
	• Rated short-circuit current	25 kA / 1 second

On-site Step-up facility

The on-site step-up facility will be built at the Project Site to connect the Monsoon wind farm to the Vietnam power grid via 500 kV double circuit 4 bundle conductors. The following key components will be installed at the on-site step-up facility.

Key components of the On-Site Step-Up Facility		
1.	115 / 500 kV Main transformer	
	• Type and configuration	Oil type transformer
	• Rated power	2 x 750 MVA
2.	500kV Transmission line facilities	
	• Transmission system type	Double circuits (ACSR4 x 330)

	• Maximum capacity	Approximate 2,000 MW (N-1)
	• Voltage	500 kV
	• Line length	Approximate 27.1 km in the territory of Lao PDR (i.e., from the Project Site to the Lao-Vietnam border)
	• Line origination	Thanh My 500 kV substation (Vietnam)
	• Line endpoint	Monsoon On-Site Step-Up Facility
	• Metering Point of Power Purchaser	At the Lao - Vietnam border (actual meters will be installed at On-Site Step-Up Facility and Thanh My Substation)
	• Tower type	Hot dip galvanized shape steel
	• Suspension tower height	Approximately 58 – 79 meters
	• Tension tower height	Approximately 54 – 66 meters
	• Foundation	Pier foundation

1.13 Project Location

The Monsoon Wind Farm Power Project is located in Dak Cheung District of Sekong Province and Sanxay District of Attapeu Province in Lao People's Democratic Republic (Lao PDR). The development also includes a 22 km 500 kilovolt (kV) transmission line, which connects to the grid in Vietnam. The project spans an area of approximately 25,000 hectares (ha).

The geographic location of the project, including the geocoordinates of the WTGs, is provided below.

Figure 1: Geographic Location of the project

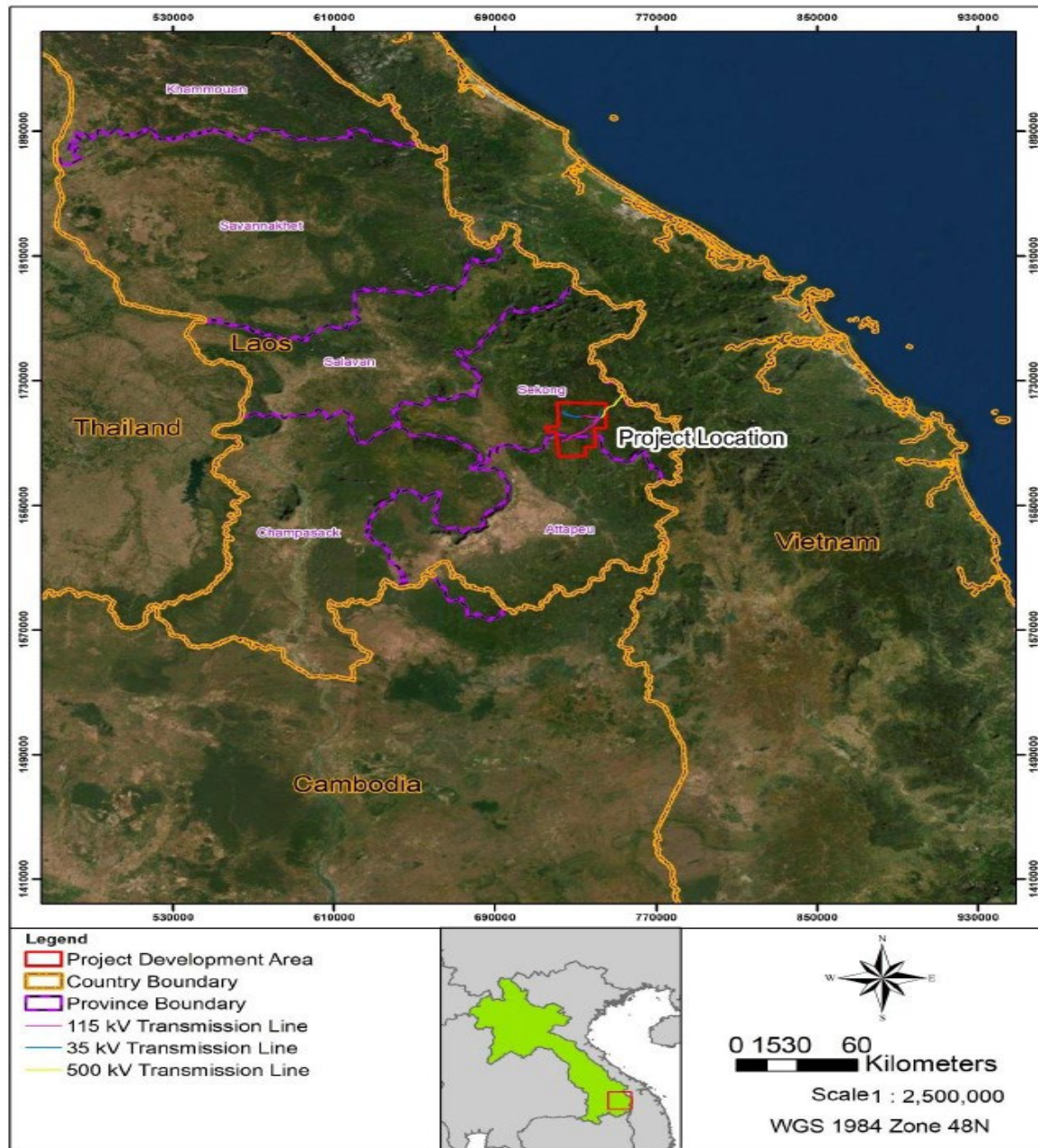
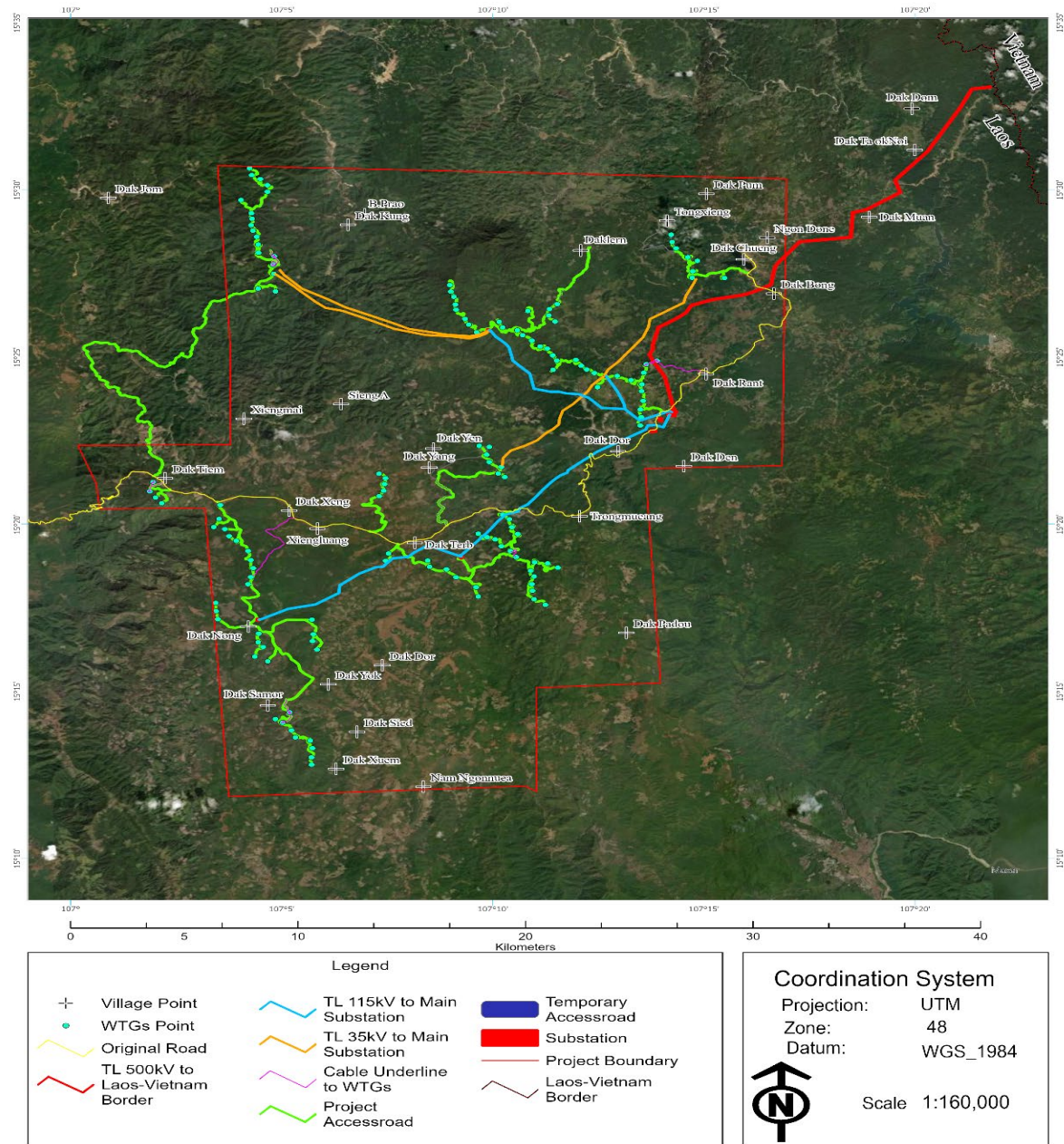


Figure-2: Geographical locations of WTGs



Co-ordinates of each WTG			
No.		Longitude	Latitude
1	WA101	107.0310	15.3496
2	WA102	107.0334	15.3466
3	WA103	107.0358	15.3437
4	WA104	106.9659	15.3426
5	WA105	106.9654	15.3369
6	WA106	106.9633	15.3326
7	WA107	107.0598	15.3308
8	WA108	107.0625	15.3274
9	WA109	107.0655	15.3249
10	WA110	107.0692	15.3200
11	WA111	107.0711	15.3152
12	WA112	107.0719	15.3115
13	WA113	107.0710	15.3069
14	WA114	107.0702	15.3034
15	WA115	107.0571	15.2942
16	WA116	107.0573	15.2900
17	WA117	107.0579	15.2859
18	WA118	107.0954	15.2857
19	WA119	107.0957	15.2783
20	WA120	107.0954	15.2748
21	WA121	107.0972	15.2706
22	WA122	107.0741	15.2787
23	WA123	107.0741	15.2745
24	WA124	107.0758	15.2720
25	WA125	107.0726	15.2671
26	WA126	107.0776	15.2648
27	WA127	107.0810	15.2361
28	WA128	107.0838	15.2343
29	WA129	107.0873	15.2305
30	WA130	107.0889	15.2271
31	WA131	107.0947	15.2255
32	WA132	107.0955	15.2214
33	WA133	107.0950	15.2168
34	WA134	107.0951	15.2132
35	WA201	107.1216	15.3585
36	WA202	107.1243	15.3564
37	WA203	107.1238	15.3532
38	WA204	107.1235	15.3499
39	WA205	107.1217	15.3471

40	WA206	107.1253	15.3187
41	WA207	107.1243	15.3149
42	WA208	107.1407	15.3152
43	WA209	107.1418	15.3120
44	WA210	107.1484	15.3110
45	WA211	107.1522	15.3067
46	WA212	107.1601	15.3021
47	WA213	107.1611	15.2971
48	WA214	107.1706	15.3308
49	WA215	107.1730	15.3358
50	WA216	107.1741	15.3317
51	WA217	107.1766	15.3285
52	WA218	107.1607	15.3287
53	WA219	107.1643	15.3268
54	WA220	107.1677	15.3256
55	WA221	107.1701	15.3224
56	WA222	107.1735	15.3210
57	WA223	107.1757	15.3171
58	WA224	107.1832	15.3144
59	WA225	107.1832	15.3144
60	WA226	107.1923	15.3115
61	WA227	107.1814	15.3101
62	WA228	107.1827	15.3073
63	WA229	107.1821	15.3036
64	WA230	107.1825	15.3001
65	WA231	107.1831	15.2958
66	WA301	107.1867	15.2930
67	WA302	107.2373	15.4741
68	WA303	107.2390	15.4707
69	WA304	107.2436	15.4663
70	WA305	107.2475	15.4648
71	WA306	107.2454	15.4593
72	WA307	107.2456	15.4561
73	WA308	107.2583	15.4561
74	WA309	107.1852	15.4219
75	WA310	107.1884	15.4208
76	WA311	107.1902	15.4176
77	WA312	107.1918	15.4150
78	WA313	107.1906	15.4105
79	WA314	107.1994	15.4153
80	WA315	107.2014	15.4124

81	WA316	107.2038	15.4103
82	WA317	107.2070	15.4086
83	WA318	107.2079	15.4055
84	WA319	107.2157	15.4071
85	WA320	107.2212	15.4053
86	WA321	107.2271	15.4129
87	WA322	107.2262	15.4089
88	WA323	107.2254	15.4029
89	WA324	107.2254	15.3998
90	WA325	107.2260	15.3947
91	WA326	107.2250	15.3903
92	WA327	107.2252	15.3868
93	WA328	107.2249	15.3824
94	WA329	107.1614	15.3725
95	WA330	107.1642	15.3681
96	WA331	107.1652	15.3644
97	WA332	107.1701	15.3620
98	WA333	107.1687	15.3585
99	WA334	107.1715	15.3569
100	WA401	107.0707	15.5107
101	WA402	107.0715	15.5072
102	WA403	107.0749	15.5051
103	WA404	107.0775	15.5024
104	WA405	107.0778	15.4991
105	WA406	107.0670	15.4947
106	WA407	107.0714	15.4885
107	WA408	107.0717	15.4855
108	WA409	107.0713	15.4824
109	WA410	107.0719	15.4795
110	WA411	107.0734	15.4767
111	WA412	107.0742	15.4727
112	WA413	107.0741	15.4685
113	WA414	107.0797	15.4626
114	WA415	107.0804	15.4583
115	WA416	107.0739	15.4508
116	WA417	107.0809	15.4492
117	WA418	107.1498	15.4527
118	WA419	107.1500	15.4499
119	WA420	107.1513	15.4468
120	WA421	107.1520	15.4432
121	WA422	107.1522	15.4400

122	WA423	107.1557	15.4359
123	WA424	107.1571	15.4329
124	WA425	107.1602	15.4289
125	WA426	107.1693	15.4341
126	WA427	107.1706	15.4313
127	WA428	107.1764	15.4299
128	WA429	107.1786	15.4275
129	WA430	107.1821	15.4250
130	WA431	107.1807	15.4398
131	WA432	107.1892	15.4407
132	WA433	107.1887	15.4375
133	WA434	107.1912	15.4342

1.14 Conditions Prior to Project Initiation

Since the project is a greenfield initiative, the pre-project conditions reflect the continuation of the existing scenario, wherein an equivalent amount of energy would have been generated by other grid-connected units. This is further elaborated in Section 3.4 (Baseline Scenario).

1.15 Compliance with Laws, Statutes and Other Regulatory Frameworks

The project is in line with all the laws and regulations of Lao PDR. The Monsoon Wind Power Project operates under a Memorandum of Understanding (MoU) between the Governments of Laos (GOL), aimed at supplying electricity to Vietnam. This collaboration ensures alignment with cross-border cooperation agreements. The project undertakes a non-recourse project financing package led by the Asian Development Bank (“ADB”) and adheres to the ADB Safeguard Policy Statement (2009) and Social Protection Strategy (2001) to maintain high environmental and social standards. Additionally, it complies with the Lao PDR Environmental Protection Law, with an Environmental and Social Impact Assessment (ESIA) approved by MONRE on July 28, 2022, addressing significant design changes and adhering to national environmental standards.

The project has secured a concession area in line with Lao Land Laws, with the concession period of 25 years from the commercial operation date. Land acquisition and compensation processes are conducted according to the Resettlement Plan and Community and Ethnic Groups Development Plan, ensuring compliance with national and ADB Safeguard Policy requirements. Furthermore, the EPC Contractor adheres to Decision No. 81/GOV (2017), which governs construction standards, environmental safety, and labor practices.

1.16 Double Counting and Participation under Other GHG Programs

1.16.1 No Double Issuance

Is the project receiving or seeking credit for reductions and removals from a project activity under another GHG program?

☐ Yes ☒ No

1.16.2 Registration in Other GHG Programs

Has the project registered under any other GHG programs?

☐ Yes ☒ No

Is the project active under the other program?

☐ Yes ☒ No

1.16.3 Projects Rejected by Other GHG Programs

Has the project been rejected by any other GHG programs?

☐ Yes ☒ No

1.17 Double Claiming, Other Forms of Credit, and Scope 3 Emissions

1.17.1 No Double Claiming with Emissions Trading Programs or Binding Emission Limits

Are project reductions and removals or project activities also included in an emissions trading program or binding emission limit? See the *VCS Program Definitions* for definitions of emissions trading program and binding emission limit.

☐ Yes ☒ No

1.17.2 No Double Claiming with Other Forms of Environmental Credit

Has the project activity sought, received, or is planning to receive credit from another GHG-related environmental credit system? See the *VCS Program Definitions* for definition of GHG-related environmental credit system.

☐ Yes ☒ No

1.17.3 Supply Chain (Scope 3) Emissions

Do the project activities specified in Section 1.12 affect the emissions footprint of any product(s) (goods or services) that are part of a supply chain?

☐ Yes ☒ No

1.18 Sustainable Development Contributions

The project supports sustainable development by generating 1,707 GWh of renewable energy annually, reducing greenhouse gas emissions, and replacing fossil fuel-based electricity. It drives economic growth through job creation, infrastructure improvements, and cross-border energy trade, enhancing energy security for Vietnam and revenues for Laos. Socially, it ensures fair compensation and community support through resettlement and engagement plans. The project aligns with Lao PDR's National Socio-Economic Development Plan and Paris Agreement commitments, with sustainability performance monitored through comprehensive Environmental and Social Management Plans.

Project Contributions to UN SDGs

	Under the Community Development Program (CDP), the Project has designated funds for a range of initiatives, including health check programs, healthcare training, scholarships, and financial assistance for vulnerable households. These efforts are aimed at enhancing the quality of life for local communities.
	A health check program for local people is being conducted for residents who lives within the project area, alongside health care training for a representative of each village. This training equips the village representatives to perform basic health checks and administer medications to the community. This initiative is conducted to ensure that the residents receive proper healthcare and to enhance health education within the community.
	Enhancing education for the talented local students is one of the objectives of the Project. As of 2024, 10 scholarships have been offered, with approximately 15 additional scholarships planed for the future. Upon graduation, the scholarship recipients will be offered the opportunity to work with the Project, ensuring job placement. In addition to this, the Project has supported the development of school infrastructure and provided vocational training for local students. These initiatives are designed to improve the quality of education and increase the number of skilled individuals within the community.
	The Project has created opportunities for women in the community to take on leadership roles within various community development programs, such as vocational training, local product development, and microfinancing initiatives. To ensure active and effective participation, female staff members of the Project also attended meetings, providing a supportive environment that encourages women to feel comfortable sharing their comments and feedback. This approach fosters greater involvement and empowerment of women in the development process.
	The Project has been focused on improving, expanding, and creating a clean water system through the installation of water treatment systems in villages within the project area. This initiative aims to enhance water quality and ensure that local residents have reliable access to clean water for drinking and daily use.

	By generating an estimated average annual output of 1,707 GWh of renewable energy, the project significantly contributes to the realization of SDG 7, which strives to ensure access to affordable, reliable, sustainable, and modern energy for all. This contribution aligns with the overarching goal of fostering a greener and more sustainable future.
	Throughout the construction and operational phases, the project has actively created numerous direct and indirect employment opportunities, predominantly benefitting the local population in the region. This proactive approach contributes to the fulfilment of SDG 8, which seeks to promote sustained, inclusive, and sustainable economic growth, along with full and productive employment and decent work for all.
	Through its efforts in emission reductions, the project aligns with the objectives of SDG 13, which calls for urgent action to combat climate change and its adverse impacts. By taking substantial measures to decrease greenhouse gas emissions, the project actively contributes to the global initiative of safeguarding the planet for the well-being of future generations.

1.19 Additional Information Relevant to the Project

1.19.1 Leakage Management

Not applicable as the project activity is not an AFOLU project.

1.19.2 Commercially Sensitive Information

No commercially sensitive information has been excluded from the public version of the project description.

1.19.3 Further Information

Not applicable, as this project activity does not involve any additional information that may have a bearing on the eligibility of the project, the GHG emission reductions, or the quantification of the project's reductions

2 SAFEGUARDS AND STAKEHOLDER ENGAGEMENT

2.1 Stakeholder Engagement and Consultation

2.1.1 Stakeholder Identification

Stakeholder Identification

The Project defines stakeholder to include workers, local communities affected by the project, national and local authorities, neighbouring projects, and/or nongovernmental organizations. The Project stakeholders were identified and prioritised by a stakeholder mapping exercise, which categorised stakeholders based on their interest in and influence over the project.

- Influence: Refers to the power stakeholders have over a project, including the ability to affect or influence decisions and facilitate its implementation.
- Interest: Refers to the priority given by the company to considering and accommodating the stakeholder's needs and interests.

The exercise determined engagement levels and tools for consultation, categorizing stakeholders as follows:

1. Quadrant 1: High influence and interest—requires proactive, regular engagement.
2. Quadrants 2 & 3: Moderate influence or interest—kept informed and consulted on relevant issues.
3. Quadrant 4: Low influence and interest—monitored and updated on key project aspects.

This categorization ensures effective stakeholder management and supports project success.

List of the stakeholders identified:

1) Project Affected Persons (PAPs)

I) Affected Population

a) Households affected by the Project's land acquisition:

- 340 HHs in Sekong
- 38 HHs in Attapeu

b) Households in 24 villages in Dak Cheung District, Sekong Province and 8 villages in Sanxay District, Attapeu province.

c) Individuals and households of nearby villages who may be make livelihood on affected lands and/or have restricted access to natural resources due to the Project footprint

- d) Individuals and households who are located within and/or nearby to the Project development area. These include:
 - o Residents of Dak Cheung District, Sekong Province
 - o Residents of Sanxay District, Attapeu Province

- e) Vulnerable groups in the affected villages

II) Cumulative Impacted Population

- a) Several villages (24 villages in Dak Cheung district of Sekong province, and 8 villages in Sanxay district of Attapeu province) likely to be affected by impacts to local livelihoods; and
- b) The administrative boundaries of Dak Cheung and Sanxay districts, as representative of all areas that could be indirectly affected by changes in ecosystem goods and services

2) Central, Provincial, District Government Agencies/Related Organisations

I) Central and Provincial Government Agencies

- a) Ministry of Natural Resources and Environment (MONRE).
- b) Provincial Department of Natural Resources and Environment (PONRE)
- c) Ministry of Energy and Mines
- d) Ministry of Planning and Investment
- e) Other relevant Ministries

II) District Administration Offices

- a) Dak Cheung District Administration Office
- b) Sanxay District Administration Office
- c) District Land Department
- d) District Agriculture and Forestry Department
- e) District Office of Natural Resources and Environment
- f) Village leaders of the 32 affected villages (including 24 villages in Dak Cheung District and 8 in Sanxay District)

3) Other Interested Parties

- I) Electricity Off taker

Vietnam Electricity (EVN)

II) Operators of meteorological radar and communication system (radio, TV, mobile telephone network sites and relay antennas)

Operators of meteorological radar and communication system online of sight of the wind turbines

III) Non-Governmental Organisations and Community Groups

a) Primarily community groups, but not limited to:

b) Youth Union

c) Lao Front for National Development

d) Lao Women Union

e) Lao Youths Revolutionary Union

f) Care International

g) Service Fraternel d'Entraide (SFE)

h) International Fund for Agricultural Development (IFAD)

i) World Food Program (WFP)

j) German-Lao Association for Development (GLAD)

k) Projahnmo Research Foundation (PRF)

l) World Wide Fund (WWF)

m) Human Rights Watch

n) BirdLife International

o) Wildlife Conservation Society (WCS)

p) International Union for Conservation of Nature (IUCN)

IV) Educational and Training Institutions (Academia)

a) Schools and training educations in Sanxay District and Dak Cheung District

V) Health Institutions

a) Community Hospital of Dak Cheung District

b) Sanxay District Hospital

c) Xieng Luang Dispensary, Dak Dor Dispensary and Dak Run Dispensary in Dak Cheung District

	d) Dak Samor Dispensary and Nam Ngon Neua Dispensary in Sanxay District VI) Financiers a) Equator Principles Financial Institutions (EPFIs) b) Asian Development Bank (ADB) c) Japan International Cooperation Agency (JICA) d) Asian Infrastructure Investment Bank (AIIB) VII) Foreign Government Multilateral Agencies a) Government of Vietnam b) International Union for Conservation of Nature (IUCN) c) World Health Organisation (WHO) d) International Labour Organization (ILO) e) The United Nations Educational, Scientific and Cultural Organization (UNESCO) VIII) Internal Stakeholders a) MWPCCL b) EPC Contractor c) O&M Contractor d) Consultants e) Workers
Legal or customary tenure/access rights	The Project signed Land Lease Agreement with GOL for both permanent and temporary land use areas. Customary practices, such as non-timber forest product (NTFP) collection and swidden cultivation, are recognized under Lao law with village certification, though communal land is not formally categorized under the Lao Land Law. Production forests provide community-level land use rights for timber and NTFP collection, with ownership retained by the state and non-transferable rights. The project design respects cultural and ceremonial sites, including cemeteries and sacred areas, to avoid significant impacts on identity-defining land uses. Customary rituals, particularly those associated with Phou Kounking Mountain, are acknowledged and integrated into planning through community consultations. These efforts align with

	both legal and customary rights, mitigating potential conflicts between stakeholders and project activities.
Stakeholder diversity and changes over time	The Monsoon Wind Power Project in Sekong and Attapeu provinces, Lao PDR involves a diverse range of stakeholders, each with distinct social, economic, and cultural characteristics. Based on the socio-economic profile, stakeholder mapping was carried out to ensure that equitable, non-discriminatory and meaningful consultation was carried out. Socially, the communities are composed of Indigenous groups Triang (75.3%, 3,973 individuals), with smaller groups including Katu (7%, 367), Yae (5.8%, 308), Arak (2.7%, 145), Lao (1.6%, 84), and a small percentage from other groups. These ethnic communities maintain distinct languages and traditions, reflecting the area's rich cultural diversity. Additionally, the stakeholders include local government authorities, NGOs, women's groups, and youth groups. Economically, the stakeholder groups have varied livelihoods, with subsistence farming, livestock raising, small business, informal labour and NTFP collection being the primary sources of income. The diverse stakeholder groups interacted in multifaceted ways, driven by their unique perspectives and interests to voice out their opinions. Overtime, the composition of these groups may undergo changes due to evolving circumstances or project related developments. New individuals or entities may join, a few might disengage. To monitor these changes and to ensure that the project continues to align with the diverse needs and expectations of its stakeholders, the project proponents will engage with the stakeholders every six months and has implemented a Grievance Redressal Mechanism (GRM).
Expected changes in well-being	The project is expected to bring significant changes to the well-being of affected stakeholders. Improved infrastructure, such as roads and connectivity, will enhance access to markets, healthcare, and education, creating new income opportunities and better living conditions. Livelihood restoration programs, including training and crop diversification, aim to address income losses from land acquisition and restricted resource access. However, economic disparities may persist, especially for vulnerable groups such as Mon-Khmer ethnicities and women. Enhanced access to healthcare and sanitation facilities is expected to improve community health,

	although increased population density from migrant workers may strain existing services. Ecosystem services critical to stakeholders, such as forests, grazing lands, and water resources, are at risk due to construction and land use changes. These resources play a vital role in livelihoods and cultural practices, and their temporary loss could impact food security and incomes. Environmental management plans, including reforestation and resource rehabilitation, aim to mitigate these impacts. Additionally, increasing connectivity and external influences may shift traditional practices and cultural norms, particularly among younger generations, requiring careful management to balance modernization with cultural preservation.
Location of stakeholders	The stakeholders, Indigenous Peoples (IPs), local communities (LCs), and customary rights holders are primarily located within the project area, encompassing Dak Cheung District in Sekong Province and Sanxay District in Attapeu Province, southern Lao PDR.
Location of resources	The Socio-Economic Survey of Affected Households (SESAH) and the Compensation Completion Report (CCR) emphasize that stakeholders, including Indigenous Peoples (IPs) and local communities (LCs), rely heavily on territories and resources in Dak Cheung District, Sekong Province, and Sanxay District, Attapeu Province. Customary agricultural lands, such as rotational upland farming plots and gardens, are essential for their livelihoods, supporting subsistence farming and ensuring food security. The CCR highlights that the project's footprint is relatively small, resulting in minimal disruption to local territories and resources. No households require resettlement or relocation, and impacts on paddy lands are minimal due to the strategic use of existing tracks and access roads.

2.1.2 Stakeholder Consultation and Ongoing Communication

Date of stakeholder consultation	The Environmental and Social Impact Assessment (ESIA, 2023) describes consultations conducted periodically since 2014, with major engagement sessions held in 2021 and 2022. These
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	sessions addressed project design updates, community impacts, and mitigation measures, ensuring meaningful input from affected communities through culturally appropriate methods. Between October and November 2021, the project conducted meetings across 23 villages in Dak Cheung District and 8 villages in Sanxay District. These included village meetings, focus group discussions (FGDs), and key informant interviews (KIs), ensuring broad community participation. In 2022, consultations continued in July and September, focusing on disclosing Environmental and Social Impact Assessment (ESIA) findings, mitigation measures, and updates to management plans. The Resettlement Plan (RP, 2023) highlights the importance of engaging with communities and authorities throughout the project lifecycle. It focuses on activities such as disclosing plans, developing livelihood restoration strategies, and fostering ongoing dialogue with stakeholders, adopting a dynamic and adaptive approach to engagement.
Stakeholder engagement process	Representatives and participants from diverse groups, such as local people, local government, women's groups, the elderly, and others, were invited to attend the consultations through public communication or phone calls. Consultation announcements, the dissemination of meeting agendas, and the Project Information Booklet were provided in advance to ensure stakeholders had sufficient information before attending the consultations. The consultations were conducted face-to-face, with attendance adjusted based on the impact category of each stakeholder. After each consultation, the project prepared a Minute of Meeting (MOM) and asked stakeholders to acknowledge and sign it for reference. The project also continued to monitor feedback and maintain ongoing engagement with stakeholders.

Consultation outcome	The Monsoon Wind Power Project conducted extensive discussions with stakeholders to ensure informed consent to project design and implementation. Through consultations outlined in the Environmental and Social Impact Assessment (ESIA) and Stakeholder Engagement Plan (SEP), key topics such as project impacts, compensation policies, and mitigation measures were addressed. Stakeholders were engaged through culturally sensitive methods, including focus group discussions, key informant interviews, and village-level meetings. While the project assessed the requirement for Free, Prior, and Informed Consent (FPIC), it concluded FPIC was not triggered as there were no significant impacts on cultural heritage or traditional land ownership. Instead, Good Faith Negotiation (GFN) principles were applied to foster transparent and participatory dialogue. Discussions on project risks, costs, and benefits highlighted concerns such as siltation of water supplies, disturbance to cemeteries, and impacts on traditional livelihoods like Non-Timber Forest Product (NTFP) collection. Stakeholders emphasized the importance of fair compensation and sustainable mitigation measures. In response, the project incorporated benefits such as scholarships, infrastructure improvements, and support for agricultural productivity. Compensation agreements were formalized with inflation-adjusted rates to ensure fairness, addressing community concerns about potential economic displacement. The project strictly adheres to Lao PDR labor laws and international standards to uphold workers' rights. Employment measures include accessible employment contracts, prevention of child labor, and social security registration for local workers. A Workers' Code of Conduct, developed in consultation with local communities, ensures respect for cultural sensitivities. Additionally, the project integrates Verified Carbon Standard (VCS) requirements into its sustainability framework, aligning climate action goals with community priorities. These efforts underscore the project's commitment to transparency, inclusivity, and regulatory compliance.
Ongoing communication	The Monsoon Wind Power Project utilizes a variety of inclusive and culturally appropriate mechanisms to maintain ongoing communication with stakeholders. Key methods include face-to-

	face meetings, workshops, and focus group discussions tailored to specific groups such as women, youth, and vulnerable populations. Written materials, including fact sheets, FAQs, and environmental and social monitoring reports, are distributed in Lao and made accessible at community locations such as village offices, schools, and health centres. Local staff facilitate communication in local dialects to ensure effective engagement with stakeholders who may have limited proficiency in the national language. Additionally, the project leverages social media platforms, such as a dedicated Facebook page, and a project website to provide updates and create channels for stakeholders to raise questions and concerns. The Grievance Redress Mechanism (GRM) plays a crucial role in the Monsoon Wind Power Project's communication strategy, offering stakeholders accessible avenues to voice concerns through feedback mailboxes, Community Liaison Officers, email, or direct visits to the project offices. Grievances are systematically tracked, evaluated, and resolved through a multi-level process involving village-level committees and the Project Grievance Redress Unit, with options to escalate unresolved issues to higher authorities if needed. The GRM promotes transparency by adhering to clear timelines, documenting outcomes, and providing regular updates to complainants. Quarterly monitoring reports detail the GRM's progress and effectiveness. Additionally, training sessions and consultations help raise community awareness about the GRM, ensuring inclusive participation. All grievance data is securely recorded in the Grievance Redress Database, enabling accountability and fostering continuous improvement in grievance management.
Stakeholder input	The project has taken due account of input received during consultations, with significant updates to project design and implementation plans made to address stakeholder concerns. For instance, the number of wind turbine generators (WTGs) was reduced from an initial 240 to 133, minimizing economic displacement and environmental impacts. This design refinement, based on stakeholder feedback and environmental studies, also reduced the infrastructure footprint, including roads and ancillary facilities, without compromising energy output.

Stakeholders raised concerns about land acquisition impacts, the preservation of cultural sites, and natural resource management. In response, the project adjusted its design to avoid culturally significant areas, such as the Phou Kounking Mountain, and implemented additional measures to protect intangible cultural heritage. Mitigation strategies for land acquisition impacts include tailored livelihood restoration plans and community development programs. These measures were developed through continuous dialogue with affected communities and local authorities.

In cases where updates to the project design were not deemed necessary, the project team provided detailed justifications. For example, when concerns about project alignment with local land use arose, studies showed that the proposed activities would not significantly affect community identity or traditional practices, ensuring compliance with local and international standards. All stakeholder feedback was documented, and the outcomes were reported back to the communities through regular updates, such as FAQs, quarterly monitoring reports, and stakeholder engagement summaries. These steps demonstrate the project's commitment to inclusive and transparent decision-making.

2.1.3 Free Prior and Informed Consent

Obtaining consent

The Monsoon Wind Power Project obtained consent to implement its activities through extensive consultations, agreements, and mechanisms tailored to the needs of Indigenous Peoples (IPs), Local Communities (LCs), and customary rights holders. Consultations included focus group discussions, key informant interviews, and village meetings, ensuring participation from all stakeholders. Villages surrounding culturally significant areas were consulted, and specific rituals were performed in collaboration with community leaders to ensure cultural sensitivity. Consent was formally documented during consultations, as seen in agreements with village leaders for activities affecting sacred sites.

Transparent agreements were reached through the Resettlement Plan (RP) and the Community and Ethnic Group Development Plan (CEGDP), which included compensation and livelihood restoration measures tailored to the needs of affected communities. Formal documentation, such as Memorandums of Understanding (MoUs),

	ensured clarity and mutual agreement on compensation terms, land acquisition processes, and other key aspects. The agreements were signed by community representatives, local authorities, and project stakeholders, ensuring compliance with legal and project standards. Ongoing conflicts are minimal, with the majority of grievances being addressed through the Grievance Redress Mechanism (GRM). Unresolved issues, such as compensation discrepancies or disputes over cultural impacts, are actively managed, and none have escalated to disrupt project activities. The GRM tracks and resolves grievances transparently, with data incorporated into monitoring reports to ensure accountability. Importantly, the project does not exacerbate existing conflicts, as demonstrated by its proactive engagement and adaptation to community feedback.
Outcome of FPIC	The Monsoon Wind Power Project conducted extensive consultations to ensure consent from Indigenous Peoples, Local Communities, and customary rights holders. Key aspects, such as project impacts, mitigation measures, and compensation plans, were disclosed transparently. The information shared included detailed asset valuations, compensation packages, and project timelines. All agreements were formalized through documented Minutes of Meetings, witnessed by local authorities, and aligned with national laws and international standards. The assessment concluded that Free, Prior, and Informed Consent (FPIC) was not applicable, as the project did not involve physical displacement, significant impacts on cultural or spiritual sites, or customary land under traditional ownership. Instead, the project adhered to principles of Good Faith Negotiation, ensuring informed consultation and participation. This approach led to project design adjustments, such as avoiding culturally sensitive areas and implementing measures to respect local customs. Additionally, livelihood restoration programs were introduced to mitigate economic impacts on communities reliant on non-timber forest products (NTFPs). There is no evidence of forced physical or economic displacement. Compensation payments were made at full replacement value before project activities commenced, ensuring compliance with international standards, including the Asian Development Bank's Safeguard Policy Statement (ADB SPS). Communities actively participated in verifying and approving compensation through

signed agreements, ensuring transparency. These efforts, supported by a robust Grievance Redress Mechanism, reinforced the project's commitment to respecting community rights and maintaining accountability throughout its implementation.

2.1.4 Grievance Redress Procedure

Development process

The grievance redress procedure for the Monsoon Wind Power Project was developed to ensure accessible, transparent, and culturally appropriate methods for addressing complaints from affected persons. Multiple channels, including feedback mailboxes, in-person reporting to Community Liaison Officers, and direct contact with the Engineering, Procurement, and Construction (EPC) contractor or Project Owner, allow stakeholders to lodge grievances easily. To accommodate cultural norms, grievances can be submitted verbally or in writing, with support provided in local languages and dialects. A dedicated Grievance Redress Mechanism (GRM) taskforce manages grievances using standardized procedures. Complaints related to construction activities, such as noise, dust, or impacts outside the compensation corridor, are handled by the EPC contractor, while broader issues are addressed by the Project Owner. All grievances are logged, tracked, and investigated promptly to ensure timely and effective resolutions.

The GRM taskforce employs culturally sensitive approaches to conflict resolution, engaging local leaders or village elders to mediate disputes where appropriate. Complainants are kept informed of the progress and outcomes at every stage, with clear timelines for resolution. Simple cases are resolved within days, while more complex grievances may require additional time, with all steps communicated transparently. If a grievance cannot be resolved at the village or project level, it is escalated to district or provincial authorities in line with Government of Laos (GoL) regulations. By integrating traditional conflict resolution practices with legal and project standards, the GRM ensures fairness, builds trust, and maintains accountability with affected communities.

Grievance redress procedure

The Grievance Redress Mechanism (GRM) comprises the following conflict resolution procedure:

Step 1: Submission of Grievance at Village Level

Affected persons can submit grievances to Community Liaison Officers, village leaders, or feedback mailboxes, either verbally or

in writing. Each grievance is screened for validity and assigned a unique tracking number. The grievance is logged, and an acknowledgment receipt is provided to the complainant. The GRM taskforce or village committee investigates the grievance and attempts resolution through mitigation or compensation measures. Urgent grievances are resolved within 7 days, while more complex cases may require additional time. All complaints and resolutions are meticulously recorded. If the grievance remains unresolved, it is escalated to the District Grievance Redress Committee (GRC).

Step 2: Escalation to District Level

If the complainant is dissatisfied with the resolution at the village level, they may escalate the grievance to the District GRC. The District GRC reviews the grievance, and if additional information is required, a field investigation is conducted. Grievances with sufficient documentation are addressed through mitigation measures or compensation. If the grievance cannot be resolved or the complainant remains dissatisfied, it is referred to the Provincial GRC.

Step 3: Escalation to Provincial Level

Unresolved grievances are forwarded to the Provincial GRC for review. The Provincial GRC discusses the case with the complainant and, if necessary, conducts additional field investigations. The committee then makes a decision and informs both the complainant and the Project Owner in writing. If the grievance is resolved, the case is closed. If unresolved, the grievance may be escalated to the judicial system.

Step 4: Escalation to Judicial Court

If a grievance remains unresolved at the provincial level, the complainant can escalate the matter to the judicial courts for legal resolution. The court's decision is final and binding. The outcome is communicated to the complainant and documented in project records.

Step 5: Engagement with Independent Accountability Mechanisms

If disputes remain unresolved, affected persons may escalate their concerns to independent mechanisms such as the Asian Development Bank's (ADB) Office of the Special Project Facilitator.

	These mechanisms provide an impartial platform for conflict resolution and additional assurance of accountability. Step 6: Recordkeeping and Reporting The GRM taskforce maintains detailed records of all grievances, including complainant details, submission date, nature of the grievance, corrective actions, and resolution outcomes. Summaries of grievances filed and resolved are included in quarterly and annual monitoring reports. These reports are made publicly available at project offices, village centers, and provincial offices as needed. The GRM ensures transparency, accountability, and adherence to both national laws and international standards throughout the grievance redress process.
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2.1.5 Public Comments

The project has not yet completed the public comment period. Once this phase is concluded, this section will be updated accordingly.

Comments received	Actions taken
NA	NA

2.2 Risks to Stakeholders and the Environment

2.2.1 Management Experience

The Monsoon Wind Power Project is a collaboration that brings together leading companies across the Asia-Pacific region to construct Southeast Asia's largest wind farm. The project is spearheaded by Impact Electronics Siam Group (IES) and involves key players of BCPG Public Company Limited (BCPG), Mitsubishi Corporation, ACEN, SMP Consultation Sole Company Limited, and STP&I Public Company Limited.

The collaboration brings a wealth of experience in renewable energy development, financing, and construction. Their combined expertise ensures the project's successful execution, from securing substantial financial backing to navigating complex regulatory frameworks and implementing cutting-edge wind technology. This collaborative effort not only advances clean energy production in the region but also fosters economic growth and strengthens regional cooperation in the renewable energy sector.

2.2.2 Risk Assessment

	Risks identified	Mitigation or preventative measure(s) taken
Natural and human-induced risks to stakeholders' wellbeing	No risk identified	No natural or human-induced permanent risks to stakeholder wellbeing has been identified. Project related activities resulting in land acquisition can have some economic impacts, which was addressed with resettlement plans where compensation was provided for loss of assets at full replacement cost, identifying land for replacement land etc., All payments including compensation for loss of land, assets, structures, trees, etc. will be made to affected households prior to physical possession
Risks to stakeholder participation	No risk identified	No risks to stakeholder participation were identified during the preparation for stakeholder consultation. The project developer analyzed relevant stakeholders, risks, opportunities, and strategies for conducting the most effective consultation. To ensure inclusiveness, the consultation venues were chosen centrally and made accessible to people of all castes, creeds, and sects in the society, ensuring that the location was considered safe by everyone in the community. Focus Group Discussions were conducted specifically to provide women and marginalized groups with an opportunity to express their opinions.
Working conditions	No risk identified	Occupational Health and Safety will be prioritized at every stage of the project. Employees will receive regular training to ensure their health and safety in the workplace. Measures will also be implemented to safeguard the health of the community near the project area. The project is committed to complying with national Occupational Health and Safety Standards, national labour standards, IFC PS2

		and International Labour Organization Labor Standards, following all guidelines and standards.
Safety of women and girls	No risk identified	The project promotes safe and secure environments for all stakeholders including women and girls. The proposed project is a renewable energy grid connected power project and therefore it's not gender sensitive and does not adversely impact the safety of women and girls
Safety of minority and marginalized groups, including children	No risk identified	The project promotes safe and secure environments for all stakeholders including minority, marginalized groups and children. The project also provides employment opportunities for local individuals, particularly from minority groups, offering positions as project staff. During public consultations with these minority groups, the project's local staff participated to ensure clear communication and to create an environment where community members felt comfortable sharing their thoughts and concerns.
Pollutants (air, noise, discharges to water, generation of waste, and release of hazardous materials and chemical pesticides and fertilizers)	No risk identified	The project is a wind energy project, hence it will not result in release of pollutants to the environment. The project will have a temporary impact on the environment only during the construction phase due to soil disruption, particulate dust generation and noise associated with vehicles, respective measures have been taken to minimize the impact. All waste from land clearing, building activities, and accommodation were collected, stored and disposed of safely in a manner that prevents any release of leachate

2.3 Respect for Human Rights and Equity

2.3.1 Labor and Work

	Risks identified ⁵	Mitigation or preventative measure(s) taken
Discrimination	No risk identified	The project is committed to ensuring a work environment free from discrimination and sexual harassment. The project developer has an Anti-sexual harassment policy in place to prohibit any form of discrimination or harassment based on factors such as race, gender, age, religion, or any other protected characteristic. All employees are made aware of these policies through training programs, and clear reporting mechanisms are in place to address any concerns or incidents promptly
Sexual harassment	No risk identified	The project is committed to ensuring a work environment free from discrimination and sexual harassment. The project developer has an Anti-sexual harassment policy in place to prohibit any form of discrimination or harassment based on factors such as race, gender, age, religion, or any other protected characteristic. All employees are made aware of these policies through training programs, and clear reporting mechanisms are in place to address any concerns or incidents promptly
Equal pay for equal work	No risk identified	The project developer has a Non-Discrimination and Equal Opportunity Policy in place, promoting equal opportunities, non-discrimination, and fair treatment of workers. The project is committed to eliminating gender-based discrimination and ensuring that everyone, regardless of gender, has equal access to employment, training, and advancement opportunities. Moreover, the project follows transparent and non-discriminatory hiring practices and has set up

⁵ The identified risks and commensurate mitigation or preventative measure(s) for forced labor, child labor, and human trafficking, must be inclusive of staff and contracted workers employed by third parties.

		a fair compensation structure based on principles of pay equity, ensuring that individuals receive equal pay for equal work.
Gender equity in labor and work	No risk identified	The project developer has a Non-Discrimination and Equal Opportunity Policy in place, promoting equal opportunities, non-discrimination, and fair treatment of workers. The project is committed to eliminating gender-based discrimination and ensuring that everyone, regardless of gender, has equal access to employment, training, and advancement opportunities. Moreover, the project follows transparent and non-discriminatory hiring practices and has set up a fair compensation structure based on principles of pay equity, ensuring that individuals receive equal pay for equal work.
Forced labor	No risk identified	Monsoon Wind Power Company Limited and its sub-contractors strictly adhere to all national and international laws related to child labor, forced labor, and human trafficking. The project unequivocally affirms that it does not and will not use victims of human trafficking, forced labor, or child labor in any aspect of its operations. This commitment aligns with the project's dedication to upholding ethical and legal standards in its workforce practices.
Child labor	No risk identified	Monsoon Wind Power Company Limited and its sub-contractors strictly adhere to all national and international laws related to child labor, forced labor, and human trafficking. The project unequivocally affirms that it does not and will not use victims of human trafficking, forced labor, or child labor in any aspect of its operations. This commitment aligns with the project's dedication to upholding ethical and legal standards in its workforce practices.

Human trafficking	No risk identified	Monsoon Wind Power Company Limited and its sub-contractors strictly adhere to all national and international laws related to child labor, forced labor, and human trafficking. The project unequivocally affirms that it does not and will not use victims of human trafficking, forced labor, or child labor in any aspect of its operations. This commitment aligns with the project's dedication to upholding ethical and legal standards in its workforce practices.
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2.3.2 Human Rights

Risks identified	Mitigation or preventative measure(s) taken
No risks identified	The project avoids any activities that could violate the economic, social, and cultural well-being of local communities. Consequently, the project developer and the project itself commit to respecting internationally proclaimed human rights. They will not be complicit in any form of violence or human rights abuse, as defined in the Universal Declaration of Human Rights. This commitment reflects the project's dedication to upholding ethical standards and safeguarding the well-being of the communities involved.

2.3.3 Indigenous Peoples and Cultural Heritage

Risks identified	Mitigation(s) or preventative measure taken
No risks identified	The Monsoon Wind Power Project implemented comprehensive consultation processes to ensure consent was obtained from Indigenous Peoples (IPs), Local Communities (LCs), and affected households (AHs). These processes adhered strictly to resettlement guidelines and international standards, ensuring that all engagement was voluntary, inclusive, and transparent. To address any potential impacts, the project provided fair compensation, livelihood restoration programs, and a Community and Ethnic Group Development Plan (CEGDP) to enhance the living conditions of affected people.

Notably, no households required physical resettlement or relocation as a result of the project. All affected households (AHs) were able to remain in their current settlements. This was achieved through careful project design and planning, which minimized the project footprint to reduce disruptions. As a result, most of the impacts were limited to non-essential areas, avoiding key cultivation lands that are vital for the livelihoods of the local communities.

The livelihood restoration and community development initiatives focused on improving the economic resilience and quality of life of AHs. These measures included training programs for alternative income generation, scholarships for education, and access to improved community infrastructure. By avoiding physical displacement and prioritizing the preservation of essential agricultural and cultural resources, the project ensured that the rights and well-being of affected communities were upheld throughout its implementation.

The Monsoon Wind Power Project implemented extensive measures to respect and protect culturally significant sites, ensuring that project activities did not disrupt sacred areas or local traditions. A key strategy was to adjust the project design to avoid culturally sensitive locations, such as sacred mountains, cemeteries, and other sites of spiritual or cultural importance. To further safeguard these areas, instruction signs and warning signs were placed to clearly demarcate the boundaries of culturally sensitive zones, ensuring that all personnel were aware of and adhered to the restrictions.

Engagement with local leaders and elders played a pivotal role in this process. Before commencing any project activities near sensitive areas, rituals and ceremonies were conducted under the guidance of community elders and spiritual leaders. These ceremonies served to honor local traditions and alleviate community concerns about potential disturbances to sacred sites.

To maintain transparency and foster trust, the project kept villages informed about the construction schedule and progress through regular updates and community meetings. This open communication allowed communities to prepare for upcoming activities and raise any concerns. Sub-contractors and workers were also provided with detailed orientation on the significance of sacred places and the importance of

respecting local customs. Training sessions emphasized appropriate behavior, cultural sensitivity, and adherence to traditional practices, ensuring that all personnel involved in the project upheld the values and traditions of the local communities.

2.3.4 Property Rights

Risks identified	Mitigation or preventative measure(s) taken
No risks identified	The Monsoon Wind Power Project implemented comprehensive consultation processes to ensure consent was obtained from Indigenous Peoples (IPs), Local Communities (LCs), and affected households (AHs). These processes adhered strictly to resettlement guidelines and international standards, ensuring that all engagement was voluntary, inclusive, and transparent. To address any potential impacts, the project provided fair compensation, livelihood restoration programs, and a Community and Ethnic Group Development Plan (CEGDP) to enhance the living conditions of affected people. Notably, no households required physical resettlement or relocation as a result of the project. All affected households (AHs) were able to remain in their current settlements. This was achieved through careful project design and planning, which minimized the project footprint to reduce disruptions. As a result, most of the impacts were limited to non-essential areas, avoiding key cultivation lands that are vital for the livelihoods of the local communities. The livelihood restoration and community development initiatives focused on improving the economic resilience and quality of life of AHs. These measures included training programs for alternative income generation, scholarships for education, and access to improved community infrastructure. By avoiding physical displacement and prioritizing the preservation of essential agricultural and cultural resources, the project ensured that the rights and well-being of affected communities were upheld throughout its implementation.

2.3.5 Benefit Sharing

As described in section 2.3.4 above, The Monsoon Wind Power Project implemented comprehensive consultation processes to ensure consent was obtained from Indigenous Peoples (IPs), Local Communities (LCs), and affected households (AHs). These processes adhered strictly

to resettlement guidelines and international standards, ensuring that all engagement was voluntary, inclusive, and transparent. To address any potential impacts, the project provided fair compensation, livelihood restoration programs, and a Community and Ethnic Group Development Plan (CEGDP) to enhance the living conditions of affected people. Notably, no households required physical resettlement or relocation as a result of the project. All affected households (AHs) were able to remain in their current settlements. Therefore, benefit sharing is not applicable and hence no benefit sharing plan has been designed.

Process used to design the benefit sharing plan	NA
Summary of the benefit sharing plan	NA
Approval and dissemination of benefit sharing plan	NA

2.4 Ecosystem Health

	Risks identified	Mitigation or preventative measure(s) taken
Impacts on biodiversity and ecosystems	The project is located within an area classified as critical habitat. In addition, the project overlaps with two Key Biodiversity Areas (KBAs), the Dak Cheung Plateau and Phou Ahyon. The Biodiversity Impact Assessment (BIA) has identified residual impacts.	The protection of natural ecosystems and biodiversity follows the mitigation hierarchy, prioritizing avoidance of disturbance and ecosystem loss. Where avoidance is not possible, impacts are minimized, rehabilitated, or offset to address any significant residual effects. Project refinements were integrated early in the planning phase to reduce environmental impacts. The turbine layout was optimized from the original 240 turbines to 148, and further refined to 133 by adopting advanced turbine

		technology that increases power generation per unit. Additionally, various re-routing and micro-siting adjustments were made during the ESIA process, including relocating the 35kV overhead line (OHL) to the south of the ridge supporting turbines, rather than both the north and south. Further minimization efforts included reducing the width of construction roads where feasible. To ensure effective mitigation, a dedicated Biodiversity Monitoring and Evaluation Plan (BMEP) has been developed, consolidating biodiversity and ecosystem service mitigation measures.
Soil degradation and soil erosion	Risk of soil erosion during construction	The project might lead to soil erosion during the construction phase. However, mitigation measures will be implemented, such as recovering topsoil stripped during clearing and construction. Land clearing will be confined to the specific project site needed for development, minimizing the impact. Furthermore, soil quality monitoring and testing will be carried out during the construction period.
Water consumption and stress	No risk identified	The project will not contaminate or impede the natural flow of any surface or sub-surface water sources. Furthermore, ground and surface water quality tests were

		conducted both during the construction and operation phase.
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2.4.1 Rare, Threatened, and Endangered Species

Is the project located in or adjacent to habitats for rare, threatened, or endangered species?

☒ Yes

☐ No

Species and habitat	The project overlaps with two Key Biodiversity Areas (KBAs), the Dak Cheung Plateau and Phou Ahyon. The Biodiversity Impact Assessment (BIA) identified residual impact on natural and critical habitats. The Ecologically Appropriate Area of Analyses (EAAAs) of volant and non-volant species contain both natural (36%-41%) and modified habitats (56%-64%) habitats in terms of the ADB SPS definitions. Furthermore, the Critical Habitat Assessment (CHA) has identified 11 species represented under Critically Endangered (CR) and Endangered (EN) status.
Areas needed for habitat connectivity	This project, overlapping with Key Biodiversity Areas (KBAs), requires achieving a Net Gain (NG) in biodiversity, aligning with IFC Performance Standard (PS) 6. To meet this requirement, the Project Proponent (PP) has developed an Initial Biodiversity Action Plan (IBAP) based on the Biodiversity Impact Assessment (BIA), outlining site-specific mitigation and offset measures to ensure a net gain scenario. A proactive approach has been adopted to anticipate, avoid, and minimize biodiversity risks and impacts: Risk Identification & Analysis – Engaging stakeholders and conducting thorough assessments to anticipate potential adverse impacts.

	• Avoidance Strategies – Exploring alternative designs, including a no-project scenario, to eliminate risks where feasible. • Impact Minimization – Reducing the project's physical footprint through optimized civil works design. • Restoration & Rehabilitation – Implementing measures such as alternative access to water sources affected by the project. Early project planning incorporated impact avoidance, leading to significant project refinements. The original turbine layout of 240 turbines was optimized to 148 and further refined to 133 by integrating advanced turbine technology, increasing energy output per turbine while minimizing land use. Additional measures, including route adjustments, micro-siting, and reducing road widths, further mitigated impacts. In line with both ADB Safeguard Policy Statement (SPS) and IFC PS requirements, biodiversity offsets are being explored to ensure no adverse effects on critical habitats and connectivity areas.
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	Risks identified	Mitigation or preventative measure(s) taken
Habitats for rare, threatened, and endangered species	Phou Ahyon, where the project's transmission line is located, is designated as a Key Biodiversity Area (KBA) and an Alliance for Zero Extinction (AZE) site. This area is home to <i>Leptobrachium xanthops</i> (yellow-eyed spadefoot toad), an Endangered (EN)	The Project has appointed a herpetologist to identify potential amphibian habitats and collaborate with project engineers to avoid these areas while developing a suitable technical layout. Additionally, the Project has raised awareness among contractors about amphibian conservation

	species listed in the IUCN database.	within the project area and has developed and implemented a Biodiversity Management Plan to monitor and ensure the effectiveness of mitigation measures.
Areas for habitat connectivity	No risk identified	<i>Leptobrachium xanthops</i> is known to inhabit a mountain stream associated with Phou Ayon Mountain, approximately 22 km north of the Monsoon Wind Farm. The planned 500 kV transmission line passes through a small portion of the Phou Ahyon KBA, traversing approximately 2.5 km of the AZE area. However, this alignment is well away from the initially recorded locations of <i>L. xanthops</i> documented in 2011.

2.4.2 Introduction of Species

This is not a AFOLU project and doesn't involve introduction of any species into the project area, hence this section is Not Applicable

Species introduced	Classification	Justification for use	Adverse effects and mitigation
NA	NA	NA	NA

Existing invasive species	Mitigation measures to prevent the spread or continued existence of invasive species
NA	NA

	Risks identified	Mitigation or preventative measure(s) taken
Invasive species	NA	NA

2.4.3 Ecosystem Conversion

The project activity has not resulted in conversion or clearing of the existing ecosystem

Risks identified		Mitigation or preventative measure(s) taken
Ecosystem conversion	NA	NA

3 APPLICATION OF METHODOLOGY

3.1 Title and Reference of Methodology

Type (methodology, tool or module).	Reference ID, if applicable	Title	Version
Methodology	ACM0002	Grid-connected electricity generation from renewable sources	22.0
Tool	TOOL07	Tool to calculate the emission factor for an electricity system	7.0
Tool	TOOL01	Tool for the demonstration and assessment of additionality	7.0.0
Tool	TOOL27	Investment analysis	14.0
Tool	TOOL24	Common Practice	3.1

3.2 Applicability of Methodology

This project is a large-scale grid-connected Greenfield renewable energy power plant. Large-scale Approved Consolidated Methodology ACM0002: Grid-connected electricity generation from renewable sources — Version 22.0⁶ is applicable for the proposed project as per the scope defined in the methodology. A detailed discussion regarding applicability of ACM0002 methodology, Version 22.0 given in table below.

Methodology ID	Applicability condition	Justification of compliance
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⁶ <https://cdm.unfccc.int/methodologies/DB/XB1TX7TAZ6SLWM9B7BC67THHVD16JV>

ACM0002	5. This methodology is applicable to grid-connected renewable energy power generation project activities that: (a) Install a Greenfield power plant; (b) Involve a capacity addition to (an) existing plant(s); (c) Involve a retrofit of (an) existing operating plant(s)/unit(s); (d) Involve a rehabilitation of (an) existing plant(s)/unit(s); or (e) Involve a replacement of (an) existing plant(s)/unit(s) or (f) Install a Greenfield power plant together with a grid-connected Greenfield pumped storage power plant. The greenfield power plant may be directly connected to the PSP or connected to the PSP through the grid.	The project activity is a Greenfield wind power plant and hence the project meets the applicability condition (a).
	7. In case the project activity involves the integration of a BESS, the methodology is applicable to grid-connected renewable energy power generation project activities that: (a) Integrate BESS with a Greenfield power plant; (b) Integrate a BESS together with implementing a capacity addition to (an) existing solar photovoltaic or wind power plant(s)/unit(s); (c) Integrate a BESS to (an) existing solar photovoltaic or wind power plant(s)/unit(s) without implementing any other changes to the existing plant(s); (d) Integrate a BESS together with implementing a retrofit of (an) existing	Not Applicable as the project doesn't involve integration of BESS

	solar photovoltaic or wind power plant(s)/unit(s). (e) Integrate a BESS together with a Greenfield power plant that is operating in coordination with a PSP. The BESS is located at site of the greenfield renewable power plant.	
	8. The methodology is applicable under the following conditions: (a) Hydro power plant/unit with or without reservoir, wind power plant/unit, geothermal power plant/unit, solar power plant/unit, wave power plant/unit or tidal power plant/unit; (b) In the case of capacity additions, retrofits, rehabilitations or replacements (except for wind, solar, wave or tidal power capacity addition projects) the existing plant/unit started commercial operation prior to the start of a minimum historical reference period of five years, used for the calculation of baseline emissions and defined in the baseline emission section, and no capacity expansion, retrofit, or rehabilitation of the plant/unit has been undertaken between the start of this minimum historical reference period and the implementation of the project activity; (c) In case of Greenfield project activities applicable under paragraph 5 (a) above, the project participants shall demonstrate that the BESS was an integral part of the design of the renewable energy project activity (e.g. by referring to feasibility studies or investment decision documents); (d) The BESS should be charged with electricity generated from the associated renewable energy power	The project involves installation of a wind power plant and is applicable under the condition (a)

	plant(s). Only during exigencies 2 may the BESS be charged with electricity from the grid or a fossil fuel electricity generator. In such cases, the corresponding GHG emissions shall be accounted for as project emissions following the requirements under section 5.4.4 below. The charging using the grid or using fossil fuel electricity generator should not amount to more than 2 per cent of the electricity generated by the project renewable energy plant during a monitoring period. During the time periods (e.g. week(s), months(s)) when the BESS consumes more than 2 per cent of the electricity for charging, the project participant shall not be entitled to issuance of the certified emission reductions for the concerned periods of the monitoring period. (e) In case the project activity involves PSP, the PSP shall utilize the electricity generated from the renewable energy power plant(s) that is operating in coordination with the PSP during pumping mode.	
	9. In case of hydro power plants, one of the following conditions shall apply: (a) The project activity is implemented in existing single or multiple reservoirs, with no change in the volume of any of the reservoirs; or (b) The project activity is implemented in existing single or multiple reservoirs, where the volume of the reservoir(s) is increased and the power density, calculated using equation (7), is greater than 4 W/m²; or	Not applicable as this project is a wind power plant

	(c) The project activity results in new single or multiple reservoirs and the power density, calculated using equation (7), is greater than 4 W/m²; or (d) The project activity is an integrated hydro power project involving multiple reservoirs, where the power density for any of the reservoirs, calculated using equation (7), is lower than or equal to 4 W/m², all of the following conditions shall apply: (i) The power density calculated using the total installed capacity of the integrated project, as per equation (8), is greater than 4 W/m²; (ii) Water flow between reservoirs is not used by any other hydropower unit which is not a part of the project activity; (iii) Installed capacity of the power plant(s) with power density lower than or equal to 4 W/m² shall be: a. Lower than or equal to 15 MW; and b. Less than 10 per cent of the total installed capacity of integrated hydro power project.	
	10. In the case of integrated hydro power projects, project participants shall: (a) Demonstrate that water flow from upstream power plants/units spill directly to the downstream reservoir and that collectively constitute to the generation capacity of the integrated hydro power project; or (b) Provide an analysis of the water balance covering the water fed to power units, with all possible combinations of reservoirs and without the construction of reservoirs. The purpose of water balance is to demonstrate the requirement of specific combination of	It's not applicable as the proposed project is a wind power project

	reservoirs constructed under CDM project activity for the optimization of power output. This demonstration has to be carried out in the specific scenario of water availability in different seasons to optimize the water flow at the inlet of power units. Therefore, this water balance will take into account seasonal flows from river, tributaries (if any), and rainfall for minimum of five years prior to the implementation of the CDM project activity.	
	11. In the case of PSP, the project participants shall demonstrate in the PDD that the project is not using water which would have been used to generate electricity in the baseline.	Not applicable as this is a wind power project
	12. The methodology is not applicable to: (a) Project activities that involve switching from fossil fuels to renewable energy sources at the site of the project activity, since in this case the baseline may be the continued use of fossil fuels at the site; (b) Biomass fired power plants/units	The proposed project is neither fuel switch nor biomass plant as it is a wind power plant. Hence, the methodology is applicable for the project activity.
	13. In the case of retrofits, rehabilitations, replacements, or capacity additions, this methodology is only applicable if the most plausible baseline scenario, as a result of the identification of baseline scenario, is “the continuation of the current situation, that is to use the power generation equipment that was already in use prior to the implementation of the project activity and undertaking business as usual maintenance”.	This condition is not applicable as the project doesn't involve any retrofits, capacity addition rehabilitations or replacements in the existing plant/unit.
In addition to satisfying the applicability criteria of the methodology ACM0002, the project also meets the applicability conditions of the tools applied as below:		

TOOL01	9. The use of the “Tool for the demonstration and assessment of additionality” is not mandatory for project participants when proposing new methodologies. Project participants may propose alternative methods to demonstrate additionality for consideration by the Executive Board. They may also submit revisions to approved methodologies using the additionality tool.	Applicable as the project participant is not using any new methodology, this tool has been applied to demonstrate additionality.
	10. Once the additional tool is included in an approved methodology, its application by project participants using this methodology is mandatory.	Applicable as the methodology used by the project requires the use of this tool.
TOOL07	3. This tool may be applied to estimate the OM, BM and/or CM when calculating baseline emissions for a project activity that substitutes grid electricity that is where a project activity supplies electricity to a grid or a project activity that results in savings of electricity that would have been provided by the grid (e.g. demand-side energy efficiency projects).	Applicable as this project is a greenfield wind power project which supplies renewable energy to the grid and this tool is used to calculate the OM, BM & CM of the project activity.
	4. Under this tool, the emission factor for the project electricity system can be calculated either for grid power plants only or, as an option, can include off-grid power plants. In the latter case, two sub-options under the step 2 of the tool are available to the project participants, i.e. option IIa and option IIb. If option IIa is chosen, the conditions specified in “Appendix 1: Procedures related to off-grid power generation” should be met. Namely, the total capacity of off-grid power plants (in MW) should be at least 10 per cent of the total capacity of grid power plants in the electricity system; or	Since the project activity is grid connected wind power project, this condition is applicable and the emission factor has been calculated accordingly.

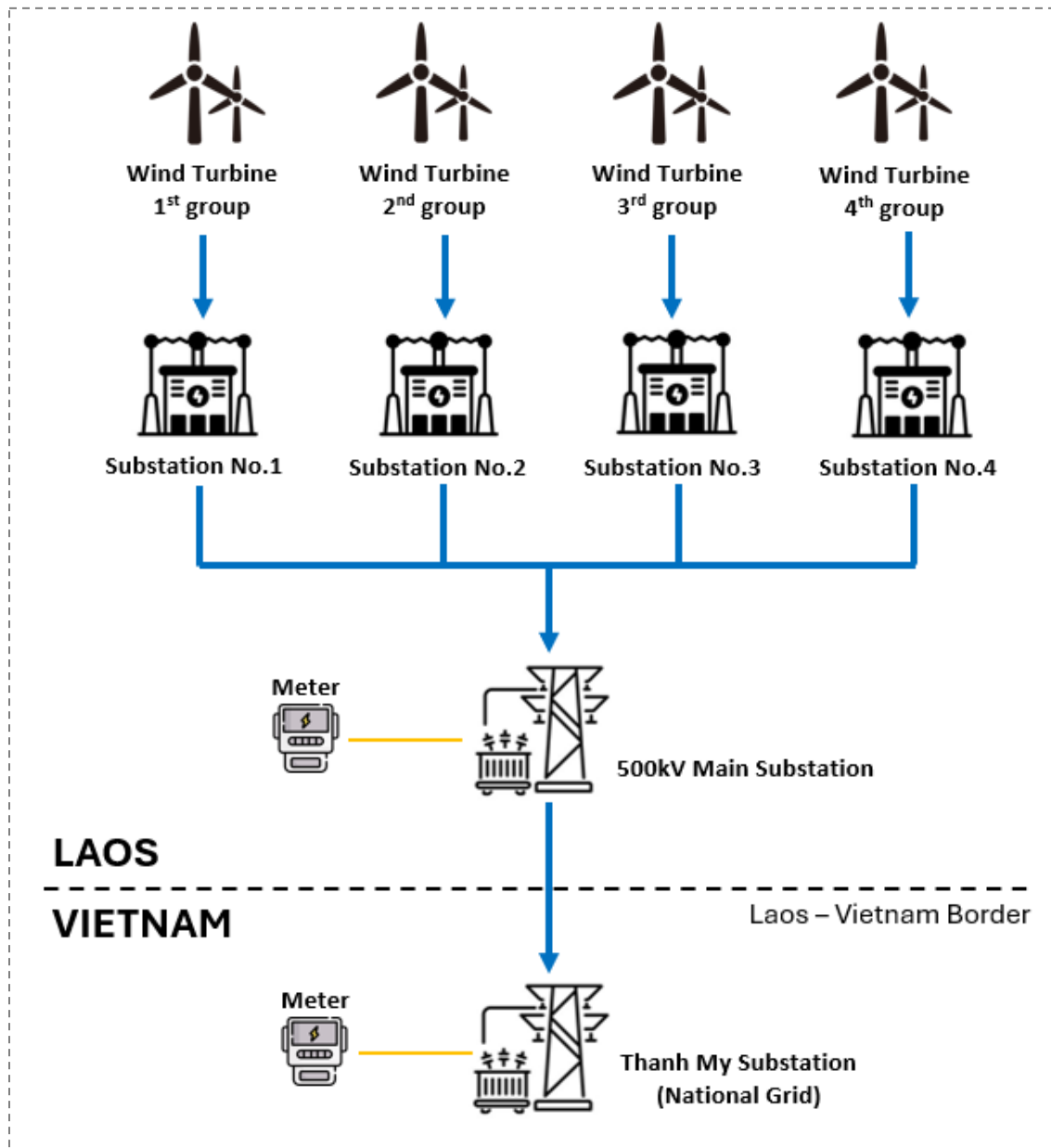
	the total electricity generation by off-grid power plants (in MWh) should be at least 10 per cent of the total electricity generation by grid power plants in the electricity system; and that factors which negatively affect the reliability and stability of the grid are primarily due to constraints in generation and not to other aspects such as transmission capacity.	
	5. In case of CDM projects the tool is not applicable if the project electricity system is located partially or totally in an Annex I country.	The project activity is located in Lao PDR, a non-Annex I country. Therefore, this criterion is not applicable
	6. Under this tool, the value applied to the CO2 emission factor of biofuels is zero	The project activity is a grid connected wind power project and therefore, this criterion is not applicable for the project activity
TOOL 24	3. This methodological tool is applicable to project activities that apply the methodological tool “Tool for the demonstration and assessment of additionality”, the methodological tool “Combined tool to identify the baseline scenario and demonstrate additionality”, or baseline and monitoring methodologies that use the common practice test for the demonstration of additionality.	Applicable as this project activity applies the methodological tool, “Tool for the demonstration and assessment of additionality.”
	4. In case the applied approved baseline and monitoring methodology defines approaches for the conduction of the common practice test that are different from those described in this methodological tool, the requirements contained in the methodology shall prevail.	The applied methodology, ACM0002 Version 22.0 requires the use of this tool to demonstrate the common practice of this project.
TOOL27	2. This methodological tool is applicable to CDM project activities and programmes of activities (PoAs) that conduct an investment analysis for the	Applicable as this project activity applies the methodological tool, “Tool for the demonstration and assessment of additionality.”

	demonstration of additionality and/or the identification of the baseline scenario.	
	3. In case the applied approved baseline and monitoring methodology contains requirements for the investment analysis that are different from those described in this methodological tool, the requirements contained in the methodology shall prevail.	The applied methodology, ACM0002 Version 22.0 requires the use of this tool to demonstrate investment analysis of this project.

3.3 Project Boundary

According to the methodology, the spatial extent of the project boundary includes the project power plant/unit and all power plants/units connected physically to the electricity system that the project power plant is connected to. Hence, the project boundary includes the project site where the power plant has been installed including wind turbines, substation, transmission line route, access road, permanent and temporary infrastructures, associated power evacuation infrastructure, energy metering points, switch yards and the connected EVN grid.

Figure-3: Project Boundary Diagram



The table below provides an overview of the emissions sources included or excluded from the project boundary for determination of baseline and project emissions:

Source		Gas	Included?	Justification/Explanation
Baseline	Source 1: CO ₂ emissions from electricity generation in fossil fuel fired power plants that are displaced due to the project activity	CO ₂	Yes	Main source of emission in the baseline
		CH ₄	No	Minor source of emission
		N ₂ O	No	Minor source of emission
		Other	No	Minor source of emission
Project	Greenfield Wind Power Project activity	CO ₂	No	As per ACM0002, for renewable energy power generation project emissions are PEY = 0
		CH ₄	No	
		N ₂ O	No	
		Other	No	

3.4 Baseline Scenario

According to ACM0002 - Version 22.0, if the project involves installing a Greenfield power plant, the baseline scenario is the electricity delivered to the grid by the project activity. This electricity would have otherwise been generated by the operation of grid-connected power plants and the addition of new generation sources, as outlined in the combined margin (CM) calculations described in "TOOL07: to calculate the emission factor for an electricity system"⁷

The combined margin ($EF_{grid,CM,y}$) represents a weighted average of two emission factors related to the electricity system: the operating margin (OM) and build margin (BM). The calculation of the combined margin must be based on data from an official source, where available, and made publicly accessible.

The combined margin of the Vietnam national grid used for the project activity is as follows:

⁷ <https://cdm.unfccc.int/methodologies/PAmethodologies/tools/am-tool-07-v7.0.pdf>

Parameter	Value (tCO ₂ /MWh)	Description	Source
EF _{grid,CM,y}	0.7896	Combined margin CO ₂ emission factor for the project electricity system in year y	Calculated as the weighted average of the operating margin (0.75) & build margin (0.25) values, sourced from data published by Department of Climate Change, "Research and develop the emission factor (EF) of Vietnam's Power Grid in 2023 (attached with CV 1726/BDKH-PTCBT)" ⁸
EF _{grid,OM,y}	0.9201	Operating margin CO ₂ emission factor for the project electricity system in year y	Calculated as the last 3 year (2021,2022 and 2023) generation-weighted average, sourced from data published by Department of Climate Change, "Research and develop the emission factor (EF) of Vietnam's Power Grid in 2023 (attached with CV 1726/BDKH-PTCBT)"
EF _{grid,BM,y}	0.3983	Build margin CO ₂ emission factor for the project electricity system in year y	Sourced from data published by Department of Climate Change, "Research and develop the emission factor (EF) of Vietnam's Power Grid in 2023 (attached with CV 1726/BDKH-PTCBT)"

3.5 Additionality

3.5.1 Regulatory Surplus

Is the project located in an UNFCCC Annex 1 or Non-Annex 1 country?

☐ Annex 1 country

☒ Non-Annex 1 country

Are the project activities mandated by any law, statute, or other regulatory framework?

☐ Yes

☒ No

If the project is located inside a Non-Annex 1 country and the project activities are mandated by a law, statute, or other regulatory framework, are such laws, statutes, or regulatory frameworks systematically enforced?

☐ Yes

☐ No

⁸ [http://dcc.gov.vn/van-ban-phap-luat/1125/Nghien-cuu,-xay-dung-he-so-phat-thai-\(EF\)-cua-luoi-dien-Viet-Nam-nam-2023-\(k%C3%A8m-CV-1726/BDKH-PTCBT\).html](http://dcc.gov.vn/van-ban-phap-luat/1125/Nghien-cuu,-xay-dung-he-so-phat-thai-(EF)-cua-luoi-dien-Viet-Nam-nam-2023-(k%C3%A8m-CV-1726/BDKH-PTCBT).html)

Not Applicable since the project activity is not mandated by any law or regulations and is entirely voluntary in nature.

3.5.2 Additionality Methods

As per ACM0002 – Version 22.0, the additionality of a project activity shall be demonstrated and assessed using the “TOOL01 - Tool for the demonstration and assessment of additionality⁹.”

The tool provides a step-wise approach to demonstrate additionality as described below:

Step 1: Identification of alternatives to the project activity consistent with current laws and regulations

Sub-step 1a: Define alternatives to the project activity

Alternatives available to the project participants or similar project developers that provide outputs or services comparable with the proposed CDM project activity. These alternatives are as below:

Alternative (a) The proposed project activity undertaken without being registered as a VCS project activity.

Alternative (c) Continuation of the current situation (no project activity or other alternatives undertaken).

Outcome of Step 1a: Alternative (a) & (c) are realistic and credible alternative scenario(s) to the project activity

Sub-step 1b: Consistency with mandatory laws and regulations

Under Lao PDR Electricity Law, there are no legal restrictions that prevent the implementation of renewable energy projects, ensuring that alternative (a) is fully in line with the country's laws and regulations.

Alternative (C), which reflects the continuation of the baseline scenario where electricity supplied to the grid by the project would have otherwise been generated by grid-connected power plants or additional generation sources, also complies with national regulations.

Outcome of Step 1b: Both alternatives (a) and (c) realistic and credible alternative scenario(s) to the project activity that are in compliance with mandatory legislation and regulations.

Step 2: Investment analysis

As per paragraph 29 of "Tool for the demonstration and assessment of additionality" version 7.0.0, it is determined that the proposed project activity is not an economically or financially

⁹ <https://cdm.unfccc.int/methodologies/PAmethodologies/tools/am-tool-01-v7.0.0.pdf>

feasible option. Referring to TOOL27 - Investment analysis, version 14.0 (EB 122 annex 01)¹⁰ for the investment analysis, the financial analysis of the project considers three options, as outlined in "Tool for the demonstration and assessment of additionality" (version 07.0.0):

Sub-step 2a: Determine appropriate analysis method:

Option I: Simple Cost Analysis

Option II: Investment Comparison Analysis

Option III: Benchmark Analysis

Since the project generates revenues from the sale of electricity, Option I is not applicable. Option II is also not applicable because there is no comparable investment alternative available to the project participant. Therefore, the most appropriate financial analysis method is Option III: the benchmark analysis, comparing returns on investment in the project activity to benchmark returns available to any investors in the country.

Sub-step 2b: Option III. Apply benchmark analysis

The project proponents have taken into account the Post-Tax Equity IRR for investment analysis during decision-making. The decision-making date is the EPC signing date which is 14th November 2022. The project proponent focuses on the returns generated on the portion of investment costs financed by them in the form of equity. According to Para 15 of TOOL 27: Investment analysis, Version 14.0, Required/expected returns on equity are suitable benchmarks for an equity IRR. Hence, the Expected return on equity is considered an appropriate benchmark. Consequently, the post-tax Equity IRR is regarded as the relevant financial indicator for Investment Analysis.

Default Value Benchmark

According to paragraph 19 of TOOL 27: Investment analysis, Version 14.0, the cost of equity is determined by choosing values provided in the Appendix. Default values for the cost of equity (expected return on equity) are presented below:

The Required return on equity (benchmark) was computed in the following manner:

$$\text{Nominal Benchmark} = \{(1 + \text{Real Benchmark}) * (1 + \text{Inflation rate})\} - 1$$

Where:

- Default value for Real Benchmark is the default value of expected return on equity in real terms for Energy Industries (Group 1) in Lao PDR as provided in the Appendix
- Average forecasted inflation rate for the host country published by the IMF for the next five years after the start of the project activity

Benchmark estimation

The Cost of Equity has been determined by using the "Methodological tool: Investment analysis" available at the time of decision-making, as well as the latest available value. As a conservative approach, the minimum value of the benchmark has been considered, calculated using these two approaches.

¹⁰ <https://cdm.unfccc.int/methodologies/PAmethodologies/tools/am-tool-27-v14.0.pdf>

Default Value at the time of investment decision:

Appendix, Investment Analysis Version 12, EB 116, Annex 2, specifies default value of expected return on equity in real terms for Energy Industries (Group 1) in Lao PDR = 20.68%¹¹

Default Value as per latest version of Investment Analysis Tool version 14.0:

Table under EB 122 annex 01 specifies default value of expected return on equity in real terms for Energy Industries (Group 1) in Lao People's Democratic Republic = 20.98%.

Thus, minimum Default Value considered for calculation of Benchmark = 20.68%

As per Para 16 of TOOL 27: Investment analysis, Version 14.0, the average forecasted inflation rate for Lao PDR published by the IMF (International Monetary Fund, World Economic Outlook) for the next five years after the start of the project activity has been used.

The five-year inflation forecast for Lao PDR as published by IMF in its World Economic Outlook Database and the corresponding benchmark value is 27.47%

Average forecasted inflation rate for 5 years	Source
27.47%	IMF, World Economic Outlook Database ¹²

The nominal benchmark has been calculated as; $\{(1+20.68\%) * (1+27.47\%)\} - 1 = 53.83\%$

Sub-step 2c: Calculation and comparison of financial indicators (only applicable to Options II and III):

Input values used in the investment analysis:

Considering the input values, Equity IRR is calculated as 9.5% against the benchmark of 53.83%. Hence, the project activity cannot be considered as financially attractive as the equity IRR for the project activity is less than the Benchmark.

Sub-step 2d: Sensitivity Analysis

In line with paragraphs 27 and 28 of TOOL 27: Investment analysis, Version 14.0, the following factors have been subjected to sensitivity analysis: 1. Tariff, 2. Generation, 3. Project Cost, and 4. O&M Cost. The rationale for sensitivity analysis is to determine the likelihood of the occurrence of a scenario other than the one presented, providing a cross-check on the suitability of the assumptions used in the development of the investment analysis.

The results of the sensitivity analysis are as follows:

The analysis reveals that even with a variation of +10% and -10% in project cost, O&M cost, generation and tariff rate, Equity IRR is significantly lower than the benchmark. As evident from the results above, the project remains additional even under the most favorable conditions.

¹¹ <https://cdm.unfccc.int/methodologies/PAmethodologies/tools/am-tool-27-v12.pdf>

¹² <https://www.imf.org/external/datamapper/PCPIPCH@WEO/LAO?zoom=LAO&highlight=LAO>

Outcome of Step 2:

This confirms that the investment is not financially attractive (Equity IRR for the project activity is less than the Benchmark Equity IRR). Thus, it can be easily concluded that the project activity is additional and is not a business-as-usual scenario.

Step 3: Barrier analysis

Barrier analysis has not been applied

Step 4: Common practice analysis

Stepwise approach for common practice analysis has been carried out as per Methodological tool “Common Practice”, version 03.1 EB84, Annex 7¹³

Definitions as per methodological tool	Justification
Step 1: calculate applicable capacity or output range as +/-50% of the total design capacity or output of the proposed project activity.	+50% of project's installed capacity: 900 MW Installed capacity of the project activity: 600 MW -50% of project's installed capacity: 300 MW
Step 2: identify similar projects (both CDM and non-CDM) which fulfil all of the following conditions: (a) The projects are located in the applicable geographical area; (b) The projects apply the same measure as the proposed project activity; (c) The projects use the same energy source/fuel and feedstock as the proposed project activity, if a technology switch measure is implemented by the proposed project activity; (d) The plants in which the projects are implemented produce goods or services with comparable quality, properties and applications areas (e.g. clinker) as the proposed project plant;	(a)The project is located in Lao PDR (b)The proposed project activity uses renewable energy to generate power. Therefore, all renewable energy projects located in Lao PDR are considered. (c)The energy source used by the project activity is Wind. Hence, only wind energy projects have been considered. (d) The project activity produces electricity; therefore, all wind power plants are considered.

¹³ <https://cdm.unfccc.int/methodologies/PAmethodologies/tools/am-tool-24-v1.pdf>

(e) The capacity or output of the projects is within the applicable capacity or output range calculated in Step 1; (f) The projects started commercial operation before the project design document (CDM-PDD) is published for global stakeholder consultation or before the start date of proposed project activity, whichever is earlier for the proposed project activity.	(e) The capacity range of the projects is within the applicable capacity range from 300 to 900 MW (f) As per CDM, for a project activity, the date on which the project participants commit to make expenditures for construction and where a contract is signed for such expenditures, the date on which the contract is signed is considered the start date and as CPA is being carried out as per the CDM tool, the EPC contract signing date i.e., 8th February 2023 is being considered as the project start date of proposed project activity. As Kyoto Protocol was ratified by Lao PDR on 6th February 2003, wind projects which had started commercial operation between 6th February 2003, 8th February 2023 have been identified. Numbers of Similar projects (CDM and non-CDM) identified based on available published sources, which fulfils the above-mentioned conditions are 0 i.e. $N_{wind} = 0$
Step 3: within the projects identified in Step 2, identify those that are neither registered CDM project activities, project activities submitted for registration, nor project activities undergoing validation. Note their number N_{all}	This is a first of it's kind project as there are no wind power projects in Lao PDR as identified in Step 2, there are no projects registered, submitted for registration nor undergoing validation Therefore $N_{all} = 0$
Step 4: within similar projects identified in Step 3, identify those that apply technologies that are different to the technology applied in the proposed project activity. Note their number N_{diff}	The identified projects in Step 3 do not apply technologies other than wind. Hence, $N_{diff} = 0$
Step 5: calculate factor $F = 1 - N_{diff}/N_{all}$ representing the share of similar projects (penetration rate of the measure/technology) using a measure/technology similar to the measure/technology used in the proposed project activity that deliver the same output or capacity as the proposed project activity.	$F = 1 - N_{diff}/N_{all}$ $F = 1 - 0/0 = 1 - 0$ Hence, F is 1
As per paragraph 18,	From the above, $F = 1$ which is greater than 0.2

The proposed project activity is a “common practice” within a sector in the applicable geographical area if the factor F is greater than 0.2 and $N_{all}-N_{diff}$ is greater than 3.	$N_{all}-N_{diff} = 0$ which is less than 3. Hence, the project activity is not a “common practice” in the applicable geographical area.
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Outcome of Step 4:

The above explanation/justification clearly shows that the proposed project activity is unlikely to be financially attractive as well as not a common practice in the host country; hence the project activity is additional.

3.6 Methodology Deviations

No methodology deviations have been applied

4 QUANTIFICATION OF ESTIMATED GHG EMISSION REDUCTIONS AND REMOVALS

4.1 Baseline Emissions

Baseline emissions include only CO₂ emissions from electricity generation in fossil fuel fired power plants that are displaced due to the project activity. The methodology ACM0002 – Ver22.0, assumes that all project electricity generation above baseline levels would have been generated by existing grid-connected power plants and the addition of new grid-connected power plants.

The baseline emissions are to be calculated as follows:

$$BE_y = EG_{PJ,y} \times EF_{grid,CM,y}$$

Where:

BE_y = Baseline emissions in year y (t CO₂/yr)

$EG_{PJ,y}$ = Quantity of net electricity generation that is produced and fed into the grid as a result of the implementation of the CDM project activity in year y (MWh/yr)

$EF_{grid,CM,y}$ = Combined margin CO₂ emission factor for grid connected power generation in year y is calculated using TOOL07 – Tool to calculate the emission factor for an electricity system, Version 07.0 (t CO₂/MWh)

Baseline emission factor is calculated as combined margin, consisting of a combination of operating margin and build margin factors according to the procedures prescribed in the latest “tool for calculating the emission factor for an electricity system”, version 07.0.

Calculation of $EG_{PJ,y}$

Since the project activity consists of the installation of new grid-connected renewable power plant at a site where no renewable power plant was operated prior to the implementation of the project activity, the project activity is the installation of a greenfield renewable energy power plant, so that:

$$EG_{PJ,y} = EG_{facility,y}$$

Where:

$EG_{PJ,y}$ = Quantity of net electricity generation that is produced and fed into the grid as a result of the implementation of the project activity in year y (MWh/yr)

$EG_{facility,y}$ = Quantity of net electricity generation supplied by the project plant/unit to the grid in year y (MWh/yr)

Calculation of $EF_{grid,CM,y}$, the grid emission factor ($EF_{grid,CM,y}$) is calculated ex-ante as per the “Tool to calculate the emission factor for an electricity-system” (Version 7.0). The emission factor is not monitored during the crediting period of the project activity but shall be updated at the renewal of the crediting period of the project activity. This methodological tool determines the CO2 emission factor for the displacement of electricity generated by power plants in an electricity system, by calculating the “combined margin” emission factor (CM) of the electricity system. The CM is the result of a weighted average of two emission factors pertaining to the electricity system: the “operating margin” (OM) and the “build margin” (BM). The operating margin is the emission factor that refers to the group of existing power plants whose current electricity generation would be affected by the project activity. The build margin is the emission factor that refers to the group of prospective power plants whose construction and future operation would be affected by the project activity.

The tool indicates six steps for the calculation of the combined margin (CM) emission factor:

STEP 1. Identify the relevant electricity systems.

For the purpose of determining the electricity emission factor, the project electricity system is defined as the electricity transmission system of Vietnam which is a single system connected by transmission lines throughout the country and owned by the Vietnam Electricity Corporation (EVN). Electricity imports from a connected electricity system are included and as per Tool to calculate emission factor of an electricity system, for the purpose of determining the operating margin emission factor, 0 tCO2/MWh is applied.

STEP 2. Choose whether to include off-grid power plants in the project electricity system (optional).

Only grid connected power plants are included in the calculation, as per Option I of the “Tool to calculate the emission factor for an electricity system” version 07.0.0

STEP 3: Select a method to determine the operating margin (OM)

The calculation of the operating margin emission factor ($EF_{grid,OM,y}$) is based on one of the following methods, which are described under Step 4:

(a) Simple OM; or

- (b) Simple adjusted OM; or
- (c) Dispatch data analysis OM; or
- (d) Average OM.

The simple OM method (Option a) is used for this study as the low-cost/must-run resources (LC/MR) constitute less than 50% of total electricity production.

As per tool to calculate emission factor for an electricity system (Version 07), The simple OM method (option a) can only be used if low-cost/must-run resources constitute less than 50% of total grid generation in: 1) average of the five most recent years, or 2) based on long-term averages for hydroelectricity production. Since the low cost/must run resources constitute less than 50% of total grid generation as seen from the average of five most recent years, the Simple OM method can be used to calculate the Operating Margin Emission factor.

Step 4: Calculate the operating margin emission factor (EF_{grid,OMSimple,y}) according to the selected method

The simple OM emission factor is calculated as the generation-weighted average CO₂ emissions per unit net electricity generation (tCO₂/MWh) of all generating power plants serving the system, not including low-cost / must-run power plants / units. The simple OM may be calculated: Option A: Based on the net electricity generation and a CO₂ emission factor of each power unit; Or Option B: Based on the total net electricity generation of all power plants serving the system and the fuel types and total fuel consumption of the project electricity system.

Department of Climate Change - Ministry of Natural Resources and Environment, "Research and develop the emission factor (EF) of Vietnam's Power Grid in 2023 (attached with CV 1726/BDKH-PTCBT)", provides information about the Combined Margin Emission Factor of Vietnam national grid. The Combined Margin is calculated ex ante using the guidelines provided by the UNFCCC in the "Tool to calculate the emission factor for an electricity system, Version 07". We have, therefore, used the Combined Margin data published in the Grid Emission Factor for GHG Reduction Project/Activity, for calculating the Baseline Emission Factor.

As per "Tool to calculate the emission factor for an electricity system", Option B ("Calculation based on total fuel consumption and electricity generation of the system") is used to calculate simple OM emission factor. Where Option B is used, the simple OM emission factor is calculated based on the net electricity supplied to the grid by all power plants serving the system, not including low cost/must run power plants/units, and based on the fuel type(s) and total fuel consumption of the project electricity system, as follows:

$$EF_{grid,OMSimple,y} = \frac{\sum_i FC_{i,y} \times NCV_{i,y} \times EFCO2,i,y}{EG_y}$$

Where:

EF_{grid,OMSimple,y} = Simple operating margin CO₂ emission factor in year y (t CO₂/MWh)

FC_{i,y} = Amount of fuel type i consumed in the project electricity system in year y (mass or volume unit)

NCV_{i,y} = Net calorific value (energy content) of fuel type i in year y (GJ/mass or volume unit)

EF_{CO2,i,y} = CO₂ emission factor of fuel type i in year y (t CO₂/GJ)

EG_y = Net electricity generated and delivered to the grid by all power sources serving the system, not including low-cost/must-run power plants/units, in year y (MWh)

i = All fuel types combusted in power sources in the project electricity system in year y

y = The relevant year as per the data vintage chosen in Step 3

Result of operating margin calculation in 2023

Year	Total power generation (MWh)	Total emissions (tCO ₂)	EF _{grid,OM simple,2023} (tCO ₂ /MWh)
	A	B	0.9201
2021	118.820.652,08	110.174.946,93	
2022	115.420.095,20	103.764.283,83	
2023	132.162.407,15	123.182.791,59	
TOTAL	366.403.154,43	337.122.022,35	

The resulting operating marginal emissions factor 2023 is 0.9201 (tCO₂/MWh).

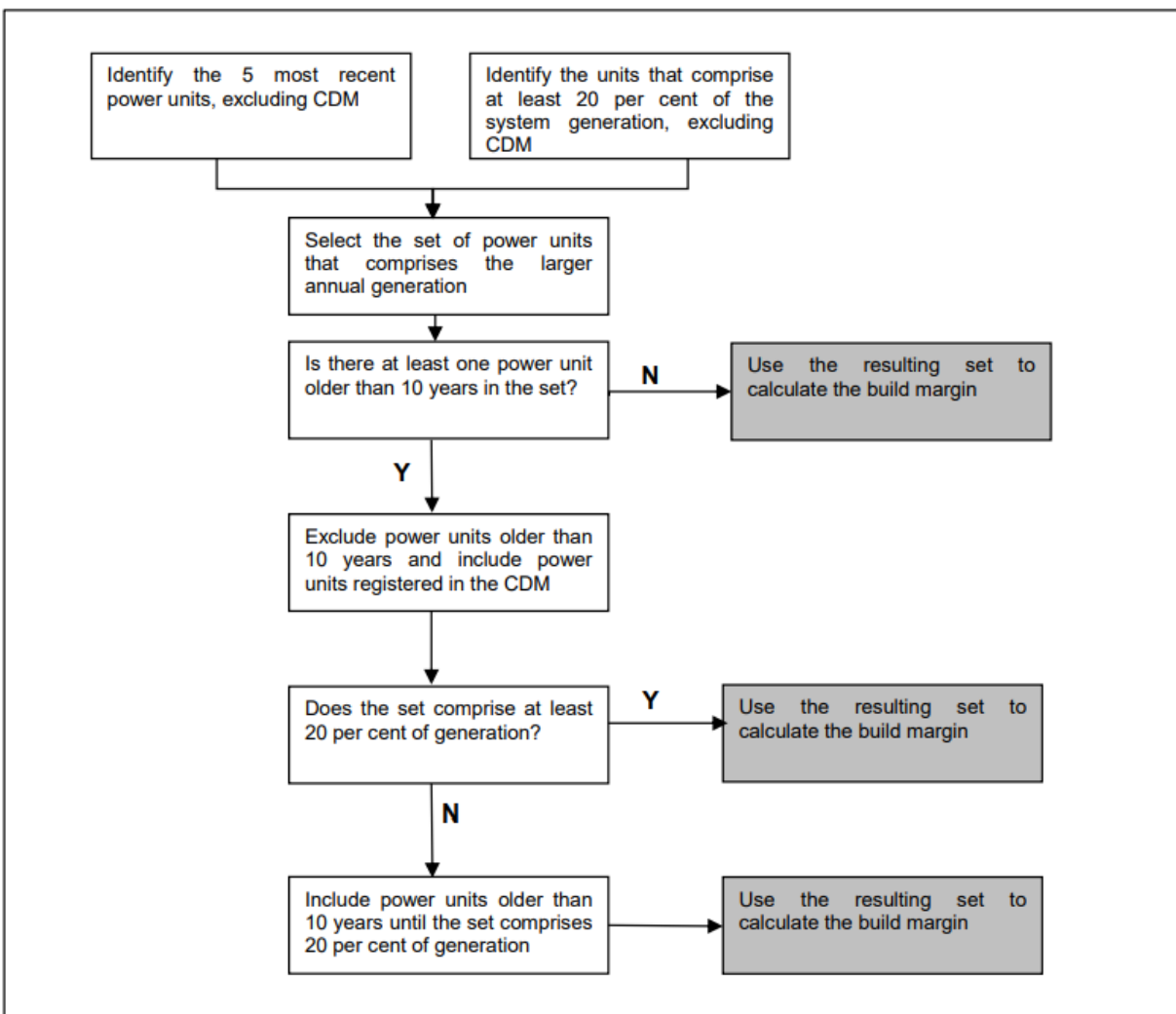
STEP 5: Calculate the build margin emission factor (EFBM,y)

The project participants have chosen Option I, i.e. fixing build margin emission factor ex ante based on the most recent information available on units already built for sample group m at the time of PSF submission to the DOE for validation. The build margin emissions factor is the generation-weighted average emission factor (tCO₂/MWh) of a sample group of power units, during the most recent year y for which power generation data is available. The Sample group of power units m used to calculate the build margin should be determined via the procedure summarized in the flow chart below:

The formula for calculating the BM emission factor is as follows:

$$EF_{grid,BM,y} = \frac{\sum m EG_{m,y} \times EF_{EL,m,y}}{\sum m EG_{m,y}}$$

Where,



$EF_{grid,BM,y}$ = Build margin CO₂ emission factor in year y (t CO₂/MWh)

$EG_{m,y}$ = Net quantity of electricity generated and delivered to the grid by power unit m in year y (MWh)

$EF_{EL,m,y}$ = CO₂ emission factor of power unit m in year y (t CO₂/MWh)

m = Power units included in the build margin

y = Most recent historical year for which electricity generation data is available

Calculation results of build margin emission factor,2023

Total electricity generation	EG 2023 (MWh)	47,402,236.48	Achieved 20.06% total electricity output in 2023
Total Emissions	EGEF 2023 (tCO2)	18.881.534,36	
Construction margin	EF _{BM,2023} (tCO2/MWh)	0.3983	

STEP 6: Calculate the combined margin (CM) emissions factor

The combined margin is the weighted average of the simple operating Margin and the build margin. In particular, for intermittent and non-dispatch able generation types such as wind and solar photovoltaic, the Tool to calculate the emission factor for an electricity system, Version 07.0.0, EB 100, Annex 4, allows to weigh the operating margin and Build margin at 75% and 25%, respectively for wind and solar projects and 50% and 50%, respectively for hydro and biomass projects. The baseline emission factor is calculated using the combined margin approach as described in the following steps:

Calculation of Baseline Emission Factor EF_y:

The baseline emission factor EF_y is calculated as the weighted average of the Operating Margin emission factor (EF_{OM,y}) and the Build Margin emission factor (EF_{BM,y}):

$$EF_{grid,CM,y} = EF_{grid,OM,y} \times w_{OM} + EF_{grid,BM,y} \times w_{BM}$$

Where,

$EF_{grid,BM,y}$ = Build margin CO₂ emission factor in year y (t CO₂/MWh)

$EF_{grid,OM,y}$ = Operating margin CO₂ emission factor in year y (t CO₂/MWh)

w_{OM} = Weighting of operating margin emissions factor (per cent)

w_{BM} = Weighting of build margin emissions factor (per cent)

The WOM and WBM weight values are applied as follows:

For wind and solar power generation projects: WOM = 0.75 and WBM = 0.25 (due to the intermittent and irregular nature) for the first credit period and for subsequent credit periods.

For all remaining projects: WOM = 0.5 and WBM = 0.5 for the first credit period and WOM = 0.75 and WBM = 0.25 for subsequent credit periods, unless otherwise specified.

Alternative weights may be proposed to suit each Country's conditions as long as WOM + WBM = 1. The values for WOM, and WBM are fixed for projects in a credit period and may be modified for subsequent credit periods.

Combined margin calculation results for 2023

For power projects using renewable energy (wind power, solar power, etc.), due to their intermittent and irregular nature, the recommended coefficients to calculate the combined marginal emission coefficient are WOM = 0.75; and WBM = 0.25 respectively. The calculation results apply to power projects using renewable energy as follows:

$$\begin{aligned} EF_{grid,CM,y} &= EF_{grid,OM,y} \times WOM + EF_{grid,BM,y} \times WBM \\ &= 0,9201 \times 0,75 + 0,3983 \times 0,25 \text{ (tCO}_2\text{/MWh)} \\ &= 0,7896 \text{ (tCO}_2\text{/MWh)} \end{aligned}$$

For other power projects, apply the marginal emission factor calculation coefficient. The recommended combinations are WOM = 0.5 and WBM = 0.5, respectively.

Based on the characteristics of Vietnam's power system, choosing the calculation coefficient of combined marginal emission factor as WOM = 0.5 and WBM = 0.5 respectively is appropriate.

The calculation results are as follows:

$$\begin{aligned} EF_{grid,CM,y} &= EF_{grid,OM,y} \times WOM + EF_{grid,BM,y} \times WBM \\ &= 0,9201 \times 0,5 + 0,3983 \times 0,5 \text{ (tCO}_2\text{/MWh)} \\ &= 0,6592 \text{ (tCO}_2\text{/MWh)} \end{aligned}$$

The baseline emission factor is ex-ante parameter and will remain constant throughout the crediting period. Thus, the combined margin grid emission factor for 2023 is 0.7896 tCO₂/MWh

4.2 Project Emissions

As per the ACM0002 ver-22.0, Project Emission for most renewable energy power generation project activities, PE_y = 0. However, some project activities may involve project emissions that can be significant. These emissions shall be accounted for as project emissions by using the following equation:

$$PE_y = PE_{FF,y} + PE_{GP,y} + PE_{HP,y} + PE_{BESS,y}$$

Where:

PE_y = Project emissions in year y (t CO₂e/yr)

PE_{FF,y} = Project emissions from fossil fuel consumption in year y (t CO₂/yr)

PE_{GP,y} = Project emissions from the operation of dry, flash steam or binary geothermal power plants in year y (t CO₂e/yr)

PE_{HP,y} = Project emissions from water reservoirs of hydro power plants in year y (t CO₂e/yr)

PE_{BESS,y} = Project emissions from charging of a BESS using electricity from the grid or from fossil fuel electricity generators (t CO₂e/yr)

The project activity involves the generation of electricity from the installation of wind projects. Hence, as per ACM0002, ver-22.0, there is no project emission for wind projects. Therefore, project emissions are zero.

4.3 Leakage Emissions

The project activity does not account for any leakage emissions. In the context of electric sector projects, potential emissions contributing to leakage, such as those from power plant construction and upstream fossil fuel use (including extraction, processing, and transport), are deemed negligible due to their limited scale and are therefore not considered in the assessment.

4.4 Estimated GHG Emission Reductions and Carbon Dioxide Removals

As per, ACM0002 ver-22.0, the ex-ante emission reductions (ER_y) for the project activity are calculated as follows:

$$ER_y = BE_y - PE_y$$

Where:

ER_y = Emission reductions in year y (tCO_{2e}/yr)

BE_y = Baseline emissions in year y (tCO_{2e}/yr)

PE_y = Project emissions in year y (tCO_{2e}/yr)

Since PE_y = 0

$$ER_y = BE_y$$

In first crediting period, the total estimated emission reduction of the project is 9,438,621tCO_{2e}, with average annual emission reductions of 1,348,374 tCO_{2e}.

Vintage period	Estimated baseline emissions (tCO _{2e})	Estimated project emissions (tCO _{2e})	Estimated leakage emissions (tCO _{2e})	Estimated reduction VCU (tCO _{2e})	Estimated removal VCU (tCO _{2e})	Estimated total VCUs (tCO _{2e})
31-Mar-2025 to 31-Dec-2025	1,019,194	0	0	1,019,194	NA	1,019,194
01-Jan-2026 to 31-Dec-2026	1,347,847	0	0	1,347,847	NA	1,347,847
01-Jan-2028 to 31-Dec-2028	1,347,847	0	0	1,347,847	NA	1,347,847
01-Jan-2029 to 31-Dec-2029	1,347,847	0	0	1,347,847	NA	1,347,847

01-Jan-2030 to 31-Dec-2030	1,347,847	0	0	1,347,847	NA	1,347,847
01-Jan - 2031 to 31-Dec-2031	1,347,847	0	0	1,347,847	NA	1,347,847
01-Jan-2032 to 30-Mar-2032	332,345	0	0	332,345	NA	332,345
Total	9,438,621	0	0	9,438,621	NA	9,438,621

For projects required to assess permanence risk:

This is not an AFOLU project and therefore is not required to assess permanence risk. Hence the below section is not applicable

i) Provide the requested information using the table below:

State the non-permanence risk rating (%)	Example: 20%
Has the non-permanence risk report been attached as either an appendix or a separate document?	☐ Yes ☐ No
For ARR and IFM projects with harvesting, state, in tCO ₂ e, the Long-term Average (LTA).	
Has the LTA been updated based on monitored data, if applicable?	☐ Yes ☐ No If no, provide justification.
State, in tCO ₂ e, the expected total GHG benefit to date.	
Is the number of GHG credits issued below the LTA?	☐ Yes ☐ No If no, provide justification.

Vintage period	Estimated baseline emissions (tCO ₂ e)	Estimated project emissions (tCO ₂ e)	Estimated leakage emissions (tCO ₂ e)	Estimated buffer pool allocation (tCO ₂ e)	Estimated reduction VCU (tCO ₂ e)	Estimated removal VCU (tCO ₂ e)	Estimated total VCU issuance (tCO ₂ e)
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DD- MMM- YYYY to 31-Dec- YYYY	Example: 50,000	Example: 20,000	Example: 10,000	Example: 4,000	Example: 8,000	Example: 8,000	Example: 16,000
01-Jan- YYYY to 31-Dec- YYYY							
01-Jan- YYYY to DD- MMM- YYYY							
Total							

5 MONITORING

5.1 Data and Parameters Available at Validation

Data / Parameter	EF _{grid,CM,y}
Data unit	tCO ₂ /MWh
Description	Combined Margin CO ₂ emission factor in year y
Source of data	Department of Climate Change - Ministry of Natural Resources and Environment, "Research and develop the emission factor (EF) of Vietnam's Power Grid in 2023 (attached with CV 1726/BDKH-PTCBT)"
Value applied	0.7896
Justification of choice of data or description of measurement methods and procedures applied	The latest EF _{grid,CM,y} for 2023 was calculated and published by Ministry of Natural Resources and Environment, Department of Climate Change (DNA of Viet Nam) on 03/12/2024 as per "Tool to calculate the emission factor for an electricity system" – Version 07.0, the calculation of the combined margin must be based on data from an official source, where available, and made publicly accessible.
Purpose of data	Calculation of baseline emissions
Comments	This parameter is fixed ex-ante for the entire crediting period

5.2 Data and Parameters Monitored

Data / Parameter	$EG_{PJ,y}$
Data unit	MWh/year
Description	Quantity of net electricity generation supplied by the project plant/unit to the grid in year y in MWh
Source of data	Monthly joint meter reading reports
Description of measurement methods and procedures to be applied	Measurements are undertaken using energy meters. The monthly net electricity supplied to the grid by the project activity is determined by subtracting the final value of exports from imports. This same value will be utilized for emission reduction calculations. All the billing meters used in the incoming feeders have an accuracy class of 0.5s.
Frequency of monitoring/recording	Continuous monitoring & monthly recording
Value applied	1,347,847
Monitoring equipment	The electricity exported / supplied by the plant to pooling substation and further to substation. This meter also measures electricity imported by the plant from the grid. The main and backup power meters ensure the provision of all necessary parameters and data of the metering systems. They determine the delivery power, monitor the operation status, and ensure the reliability of the metering systems. These meters automatically record and save values, capturing power values for two-way delivery at 24 hours on the last day of the billing period.
QA/QC procedures to be applied	The meters are approved, tested & sealed by the State Utility. The meters are in the custody of State Utility. The frequency of calibration is once in 5 years. The monthly electricity supplied/exported by the project activity in the JMR report is cross checked with the monthly invoices of sale. In the absence or delay in the meter calibration appropriate guidelines will be applied appropriately to confirm the conservativeness of metering. The metering arrangement, accuracy class of meters, calibration frequency is under control of state electricity board and PP does not have any control on it. PP is getting value of net electricity supplied to grid and the same is considered the monitoring parameter. The billing is raised based on substation meters.

Purpose of data	Calculation of baseline emissions
Calculation method	Net electricity supplied to the grid by the project plant in a given month = Export(kWh)– Import (kWh)
Comments	Data will be archived in paper & electronic form for two years after the end of crediting period or of the last issuance of VERs for this project activity, whichever occurs later.

5.3 Monitoring Plan

The monitoring plan aims to make sure the project accurately measures emission reductions during its crediting period. The project owner, with support from EVN, will carry out this plan. It ensures that actual, measurable, and long-term reductions in greenhouse gas (GHG) emissions are well-tracked and reported, playing a vital role in determining the project's overall emission reductions.

The equipment of the power metering system ensures compliance with technical and management requirements in power delivery as stated in GOV regulations and EVN's regulations.

The metering system is as below:

- Main measuring point

- At the 500kV feeder bays of 500kV Thanh My substation which is connected to 500kV Monsoon substation;
- At the main measuring point there is one installed power metering system which has: current and voltage transformer core; meter with precision level 0.2.

- 1st backup measuring point

- At the 500kV feeder bays of 500kV Monsoon substation which is connected to 500kV Thanh My substation;
- At the 1st backup measuring point there is one installed power metering system which has: current and voltage transformer core; meter with precision level 0.5.

- 2nd backup measuring point

- At the 500kV transformer terminals of 220kV Monsoon substation;
- At the 2nd backup measuring point there is one installed power metering system which has: current and voltage transformer core; meter with precision level 0.5.

The steps of monitoring the electricity supplied to the grid and the electricity imported from grid and consumed by the proposed project are as follows:

1) The electricity supplied by the project to the grid and electricity imported from the grid are measured automatically by the bi-directional meter systems (main and backup power meters). The data is measured continuously;

2) Persons in charge of data record and meter supervisor from Monsoon Wind Power Plant together with staff from EVN shall read and collect data from main power meters on the first day of every month, the result (monthly electricity protocol) will be signed by both parties and kept respectively;

3) The data from the backup power meters will be cross checked with the data from main power meter. The data from back-up system will be used in case of failing of the main meter;

- 4) The project owner provides electricity sales invoice to EVN, and keeps the copy of invoices;
- 5) The power owner provides the electricity protocols and copy of invoices and other related documents to the verifier of VVB.

APPENDIX 1: COMMERCIALLY SENSITIVE INFORMATION

Section	Information	Justification
	NA	NA

APPENDIX X: <TITLE OF APPENDIX>

Use appendices for supporting information. Delete this appendix (title and instructions) where no appendix is required.